# Physics-Informed Curriculum Learning Framework for Hyperspectral Underwater Target Characterization

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***Abstract*—Hyperspectral image (HSI) provides fine-grained spectral information essential for material identification and target detection, particularly in complex environments, such as underwater scenarios. However, hyperspectral underwater target detection (HUTD) remains challenging due to severe spectral distortions and variability introduced by wavelength-dependent absorption and the dynamic nature of aquatic environments. Existing separation-based and characterization-based methods are often constrained by weak signal responses or a heavy reliance on accurate environmental parameter estimation, which is difficult to achieve in practice. To overcome these limitations, we propose PCL-HUTD, a novel physics-informed curriculum learning (PCL) framework for robust underwater target characterization without requiring explicit environmental modeling. PCL-HUTD integrates a physics-guided target construction module with a hard-sample aware contrastive learning strategy, enhanced by unsupervised clustering and a perturbation-consistency-based sample selection mechanism. Furthermore, a closed-loop CL paradigm is introduced to progressively refine target representations throughout training. Extensive experiments on three real-world HUTD datasets demonstrate that PCL-HUTD achieves state-of-the-art (SOTA) performance in both detection accuracy and robustness, particularly under challenging conditions with strong background interference. These results validate the effectiveness of our parameter-free, physics-informed approach for underwater hyperspectral target detection (HTD).**

***Index Terms*—Contrastive learning, curriculum learning (CL), hyperspectral imaging, physics-informed modeling, spectral variability, target characterization, underwater target detection.**

## I. Introduction

HYPERSPECTRAL imaging [1], [2], [3], [4], [5], [6] has emerged as a fundamental technique in remote sensing, capable of acquiring continuous spectral information across hundreds of narrow and contiguous bands with high fidelity. The resulting rich spectral signatures enable precise material characterization and reliable target detection [7], [8], [9], [10], [11], [12], [13], particularly under complex environmental conditions [14]. Among such conditions, underwater scenarios present a representative and challenging case due to strong wavelength-dependent absorption, which significantly degrades the performance of conventional optical sensors. Specifically, water exhibits strong absorption near 0.45 (blue) and 0.65 $\mu$m (red) [15], resulting in substantial information loss within these spectral regions. As RGB and multispectral sensors primarily rely on limited spectral bands, their ability to extract informative cues for underwater target detection is inherently constrained, thereby resulting in suboptimal detection performance. In contrast, hyperspectral images (HSIs), with their broad spectral range and fine spectral resolution, can capture wavebands less affected by water absorption. This capability makes HSI a promising solution for underwater target detection. Despite the advantages of hyperspectral imaging, two major challenges remain. The first challenge lies in the spectral mismatch between the reference and observed spectra. Conventional hyperspectral detection typically assumes that a laboratory-measured reference spectrum accurately represents the target. However, in underwater environments, wavelength-dependent attenuation significantly distorts spectral signatures, resulting in a substantial deviation between the reference and the observed spectra. For instance, as illustrated in Fig. 1(a), the spectrum captured in a coastal water environment differs markedly from its laboratory counterpart. Consequently, the reference spectrum can no longer serve as a reliable representation of the target, thereby complicating the detection process. The second challenge is the high level of spectral variability in underwater environments. Spectral variability

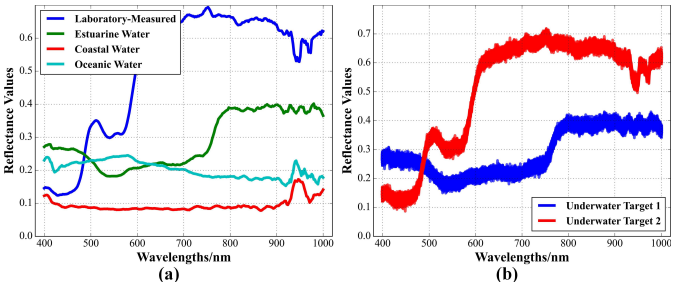



Fig. 1. Illustration of the challenges in HUTD task. (a) Spectral mismatch between the reference and observed spectra caused by water absorption under different underwater conditions. (b) Significant spectral variability induced by the dynamic and complex nature of nearshore environments.

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refers to changes in the spectral signature of a target that result from differences in lighting or sensor viewing angles, and it is a known issue in hyperspectral detection on land [16]. In the underwater scenes, this problem becomes more serious because of the complex and nonlinear interaction between light and the water medium, which leads to much higher uncertainty in the observed spectra, as illustrated in Fig. 1(b). These two problems form the core challenges of underwater hyperspectral target detection (HTD) and have motivated two mainstream solution paradigms in recent research.

*Separation-based methods* aim to reduce the effect of water attenuation by treating it as an interfering signal that is mixed with the target response. This approach attempts to recover the original target spectrum from the observed underwater spectrum through signal separation. Qi et al. [17] proposed the first separation-based method for hyperspectral underwater target detection (HUTD) by modeling the observed spectrum as a linear combination of the target and water spectra. They used hyperspectral unmixing techniques to isolate the pure target spectrum for detection. Liu et al. [18] extended this idea by applying nonlinear representations in order to better describe the spectral complexity of nearshore environments. More recently, Li et al. [19] further improved detection performance by incorporating pure endmember information and abundance estimation through a weighting and spectral fusion process. Although these methods have shown progress, they are fundamentally limited by hyperspectral unmixing, which often treats weak signals as noise and removes them during the separation process. This limitation is especially critical in remote sensing scenarios, where targets often occupy only a few pixels, leading to what is commonly referred to as the weak signal problem [20]. In the underwater environments, this issue becomes even more severe due to strong spectral attenuation, which further reduces signal strength. Thus, the separation-based methods often face difficulties when applied to underwater scenes with complex background interference.

To address the weak signal problem, *characterization-based methods* aim to directly model the observed underwater spectrum under given environmental conditions. These methods support target detection without relying on signal strength. They describe the transformation from the reference spectrum to the observed spectrum using physical models of underwater imaging that include specific parameters, such as water depth and water composition [21], [22], [23]. For example, Jay et al. [21] proposed a method based on the generalized likelihood ratio test, which uses a bathymetric filter to estimate unknown water properties and detect targets at the same time. Gillis [22] later introduced a physics-based framework that considers spectral variability caused by environmental changes and applies nonlinear dimensionality reduction to support underwater target detection. More recently, Qi et al. [23] developed a self-improving framework that refines both the estimation of environmental parameters and the detection of targets by using spectral and spatial cues. These methods have shown considerable performance in various underwater scenes, even in the presence of heavy background interference. However, their effectiveness depends heavily on the accuracy of environmental parameter estimation. In practice, such estimation is often unreliable due to the complexity and variability of underwater conditions [24], [25]. Besides, the estimation process usually requires high computational cost because of iterative optimization over high-dimensional HSIs.

These limitations raise an important question: *whether it is possible to develop a target characterization method that does not require explicit parameter estimation*. To explore this idea, we propose a physics-informed curriculum learning framework for HUTD, called PCL-HUTD, which combines the physical principles of underwater imaging with a progressive learning strategy. This framework enables robust hyperspectral underwater target characterization without the need for explicit environmental parameter estimation.

In particular, PCL-HUTD includes a physics-guided target construction module that is designed based on the hyperspectral underwater image formation model. By embedding the physical degradation process into the structure of the network, this module improves the accuracy and interpretability of underwater target spectrum construction compared to conventional data-driven models. To guide this construction process, we introduce a contrastive learning strategy that focuses on difficult examples. This strategy encourages the constructed target representations to align more closely with potential underwater spectra, while enhancing their ability to be distinguished from background signals. Since contrastive learning requires the identification of target pixels as positive samples and background pixels as negative samples, we use an unsupervised clustering method to infer the semantic category of each pixel. To improve clustering reliability, we use intermediate outputs from the target construction module as prior cues and apply a perturbation-consistency strategy to select high-confidence samples using a predefined threshold.

Moreover, the interaction between the target construction and clustering processes forms a closed-loop mechanism. The clustering process benefits from physically guided target information, while the target construction is improved by using the most reliable samples selected through clustering. To enhance this mutual reinforcement, we design a curriculum learning (CL) strategy that gradually increases the number of selected reliable samples. This approach allows the model to start with simpler cases and gradually incorporate more complex ones, which helps to ensure stable convergence and improve overall robustness. After training is completed, the final output of the target construction module is used as the result for underwater target characterization. This result can be directly applied as a reference spectrum for subsequent target detection.

To evaluate the effectiveness of the proposed PCL-HUTD framework, we conduct comprehensive experiments on three real-world HUTD datasets. The experimental results show that PCL-HUTD consistently outperforms existing state-of-the-art (SOTA) methods in both detection accuracy and robustness. This advantage is more evident under challenging underwater conditions with strong background interference. The main contributions of this work are as follows.

1) We propose PCL-HUTD, a novel framework that enables parameter-free underwater target characterization by integrating physics-informed modeling with

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curriculum-driven learning, offering improved adaptability to diverse underwater conditions.
2) We design a physics-guided target construction module, which embeds the underwater image formation model into network architecture to enhance the interpretability and fidelity of spectral reconstruction.
3) We introduce a hard-sample aware contrastive learning strategy with unsupervised clustering and perturbation-consistency selection, enabling accurate and label-free discrimination between target and background pixels.
4) We implement a closed-loop CL framework that allows mutual refinement between representation construction and clustering reliability, progressively improving robustness through adaptive sample inclusion.
5) Extensive experiments on real-world HUTD datasets demonstrate that PCL-HUTD consistently outperforms SOTA methods in terms of detection accuracy and robustness, particularly under strong background interference.

## II. Related Work

### A. Hyperspectral Target Characterization

Target characterization is a fundamental component of HTD, aiming to construct a reference spectrum that accurately represents the target of interest. This reference spectrum serves as the basis for detection and is critical for distinguishing the target from complex backgrounds in hyperspectral scenes.

One of the earliest representative works, Ma and Du [26], enhanced HTD by incorporating region-based background modeling into the matched filter (MF) framework, enabling more effective utilization of the reference spectrum. Building on this, Zare et al. [27] introduced a multiple instance learning (MIL) framework that estimates discriminative target signatures directly from imprecisely labeled training data, thereby eliminating the need for pure pixels or precise annotations. Zhong et al. [28] further extended this approach to multitarget scenarios by adopting a diversity-promoting learning strategy to characterize multiple target spectra. More recently, Jiao et al. [29] applied the hyperspectral characterization framework to subpixel detection by integrating a linear spectral mixing model, thus extending its applicability to more realistic and complex detection settings.

In this work, we focus on a more challenging task: underwater target characterization. Compared with terrestrial scenarios, underwater environments introduce severe spectral distortion due to the complex and changing optical properties of water, which greatly complicates the characterization process. In addition, unlike terrestrial scenes where rough spatial annotations are often available, underwater environments are more challenging because water absorption and scattering reduce spatial details. Thus, the characterization process must rely on an inaccurate reference spectrum that deviates from the true underwater response, further increasing the difficulty of accurate target representation.

### B. Hyperspectral Underwater Imaging

Hyperspectral underwater imaging (HUI) refers to the process of acquiring and analyzing hyperspectral data in underwater environments, and it provides the theoretical foundation for this work. HUI models the formation of HSIs by characterizing the interaction between light, the water medium, and submerged objects. A widely adopted formulation is based on a bathymetric model [30], [31], which describes the observed spectrum $\boldsymbol{r}(\lambda)$ at wavelength $\lambda$ as

$$\boldsymbol{r}(\lambda) = \boldsymbol{r}_w(\lambda)\left[1 - e^{-\boldsymbol{K}(\lambda)H}\right] + \boldsymbol{r}_t(\lambda)\, e^{-\boldsymbol{K}(\lambda)H} \tag{1}$$

where $\boldsymbol{r}_w(\lambda)$ is the spectrum of the water body, $\boldsymbol{K}(\lambda)$ represents the wavelength-dependent attenuation coefficient, $H$ denotes the depth of the target, and $\boldsymbol{r}_t(\lambda)$ corresponds to the pure target spectrum. This model captures two fundamental physical characteristics of underwater spectral formation.

1) The observed spectrum $\boldsymbol{r}(\lambda)$ is a weighted combination of the water and target spectra, reflecting their mixed contributions at different depths and wavelengths.
2) The combination weights are governed by exponential decay, which implicitly satisfies the sum-to-one constraint under energy conservation.

These physical properties form the basis for the design of our physics-informed target construction module, which serves as the core of the proposed PCL-HUTD framework. By embedding this formulation into the learning architecture, the model can incorporate domain knowledge to achieve more interpretable and reliable underwater target characterization.

## III. Motivations

This section presents the core motivation of PCL-HUTD. PCL-HUTD is specifically designed to address the challenges of HUTD by adhering to *three key design principles*: contrastive target characterization, consistency-guided reliable sample selection, and progressive CL.

### A. Contrastive Target Characterization

Target characterization is a common technique in HUTD, where existing methods aim to predict the underwater target spectrum using estimated environmental parameters and the reference spectra based on the bathymetric model, as illustrated in Fig. 2(a). Despite recent progress, these methods face two major limitations.

1) *Characterization errors due to inaccurate parameter estimation:* Fig. 2(a) shows that the characterization results are highly sensitive to the estimated environmental parameters. However, existing estimation strategies [24], [25] often struggle to produce accurate results, particularly in complex underwater scenes with strong background interference. This leads to *error accumulation* during the characterization process, causing significant deviations between the characterized spectra and the actual underwater target spectra.
2) *Characterization errors caused by invalid model assumptions:* The bathymetric model assumes a linear underwater imaging process and spectral independence between the target and background. These assumptions may not hold in real-world environments [18], causing discrepancies between the theoretical model and actual

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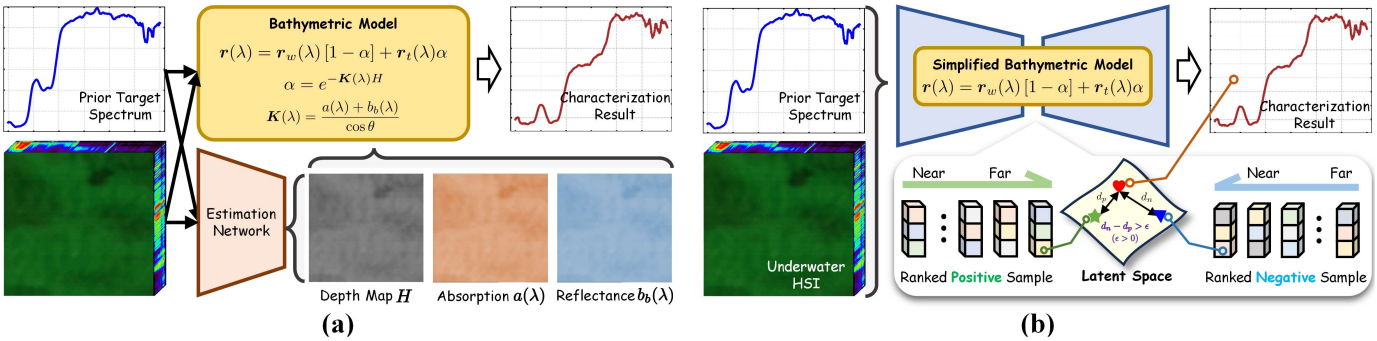



Fig. 2. Comparison between (a) traditional target characterization approach and (b) proposed contrastive target characterization pipeline. The latter focuses on discriminability, rather than absolute accuracy, leveraging a hard-aware contrastive learning strategy to enhance TBS.

imaging conditions, and further degrading characterization accuracy.

These limitations suggest that the core issue of the existing methods lies in the fixed and overly literal characterization perspective, which seeks to *approximate the true underwater target spectrum as closely as possible*. However, for the HUTD task, the goal is not to recover the exact spectrum, but rather to generate a spectral representation that effectively distinguishes the target from the background. From this viewpoint, strictly matching the real underwater target spectrum is unnecessary.

Motivated by this insight, we propose a novel *contrastive target characterization pipeline*, which *prioritizes discriminability over absolute accuracy*. As shown in Fig. 2(b), the bathymetric model in (1) is used only to provide physical interpretability. Instead of relying on estimated parameters, the coefficients for constructing the characterized target spectrum are learned through an optimization process defined by a proxy task. This optimization maximizes the spectral distance between the characterized target and background samples while minimizing the distance to potential target spectra. To further enhance discriminability, we introduce a hard-example aware ranking strategy that focuses on difficult target and background cases, encouraging the model to better separate target and background signals. Finally, to address the limitations of the bathymetric model's linear assumption, the optimization is carried out in a latent space, rather than in the original spectral space. This design leverages the expressive power of deep neural networks to model complex nonlinear relationships. This allows the model to learn more flexible and robust representations that better capture the complexities of underwater imaging, thereby improving target-background separability (TBS) in the characterized spectrum.

### B. Consistency-Guided Reliable Sample Selection

As described in Section III-A, contrastive learning depends on reliable identification of positive (target) and negative (background) samples [32], [33]. However, in practical HUTD settings, such pixel-level annotations are not available, making it difficult to construct trustworthy contrastive pairs.

To address this, we use *unsupervised clustering* to group pixels and infer semantic categories. Cluster centroids are compared with the characterized target spectrum to determine candidate labels. Ideally, accurate clustering would allow for effective sample selection, thereby improving the characterization performance. In practice, however, incorrect assignment of positive or negative labels can do more harm than good. Thus, a refinement mechanism is needed to retain only the most reliable samples.

Following the principle that high-quality clusters should exhibit considerable *compactness* and *stability* [34], we propose a *consistency-guided sample selection strategy*. As shown in Fig. 3(a), when reclustering with a stricter criterion, excellent clusters preserve their original members, while poor ones lose samples to the unclustered set. Conversely, as shown in Fig. 3(b), under a looser criterion, excellent clusters still maintain their composition, whereas poor clusters tend to absorb samples from other clusters or from the unclustered set. These observations indicate that *the consistency of clustering results under different criteria can be used as an indicator of sample reliability*. Motivated by this, we propose a consistency-guided reliable sample selection strategy under clustering perturbations in Fig. 4.

Specifically, we perform clustering under three settings: a normal criterion, a stricter one, and a looser one. Due to the changes in cluster membership and centroid locations under different settings, the resulting cluster labels are not directly aligned. Therefore, we apply a posthoc alignment procedure (e.g., centroid matching or label permutation) to enable consistent comparison. Finally, we retain only those samples that remain consistently assigned to the same cluster under both perturbed criteria and discard the rest as unreliable. This approach effectively filters out unstable samples, improving the robustness of the contrastive target characterization.

### C. CL Manner

In addition to the clustering-guided sample selection, the overall training process of PCL-HUTD is further enhanced through a *CL strategy* that gradually incorporates more complex examples. This design is consistent with the mutual refinement mechanism between the target construction module and the clustering module, where each component iteratively benefits from the output of the other.

At the beginning of training, the model focuses on a limited set of *high-confidence samples*, which are selected based on their clustering consistency under perturbations (as described in Section III-B). These samples provide a stable foundation

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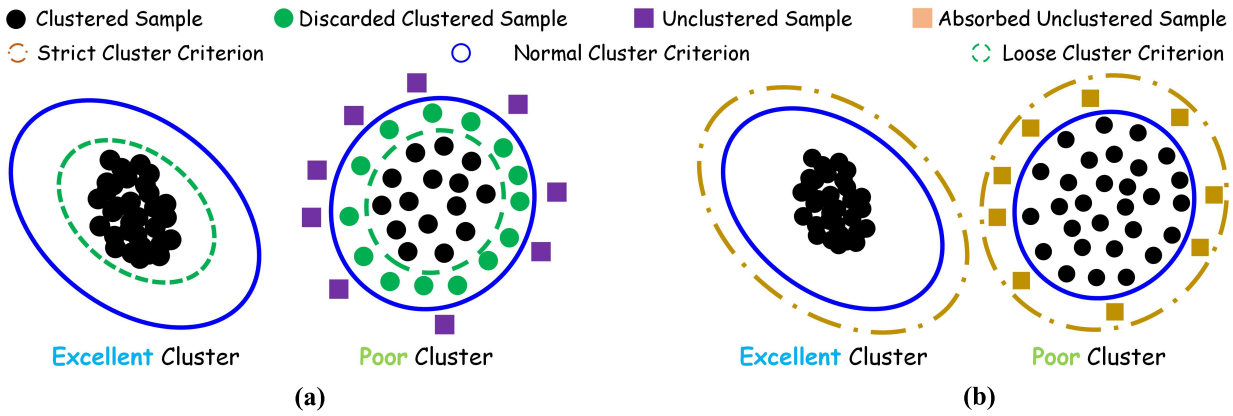



Fig. 3. Illustration of the motivation for consistency-guided reliable sample selection. (a) Clustering result underwater normal and stricter criteria, where excellent cluster preserves its original members and poor one lose samples to the unclustered set. (b) Clustering result under normal and looser criteria, where excellent cluster maintain their composition and poor one tends to absorb samples from the unclustered set.

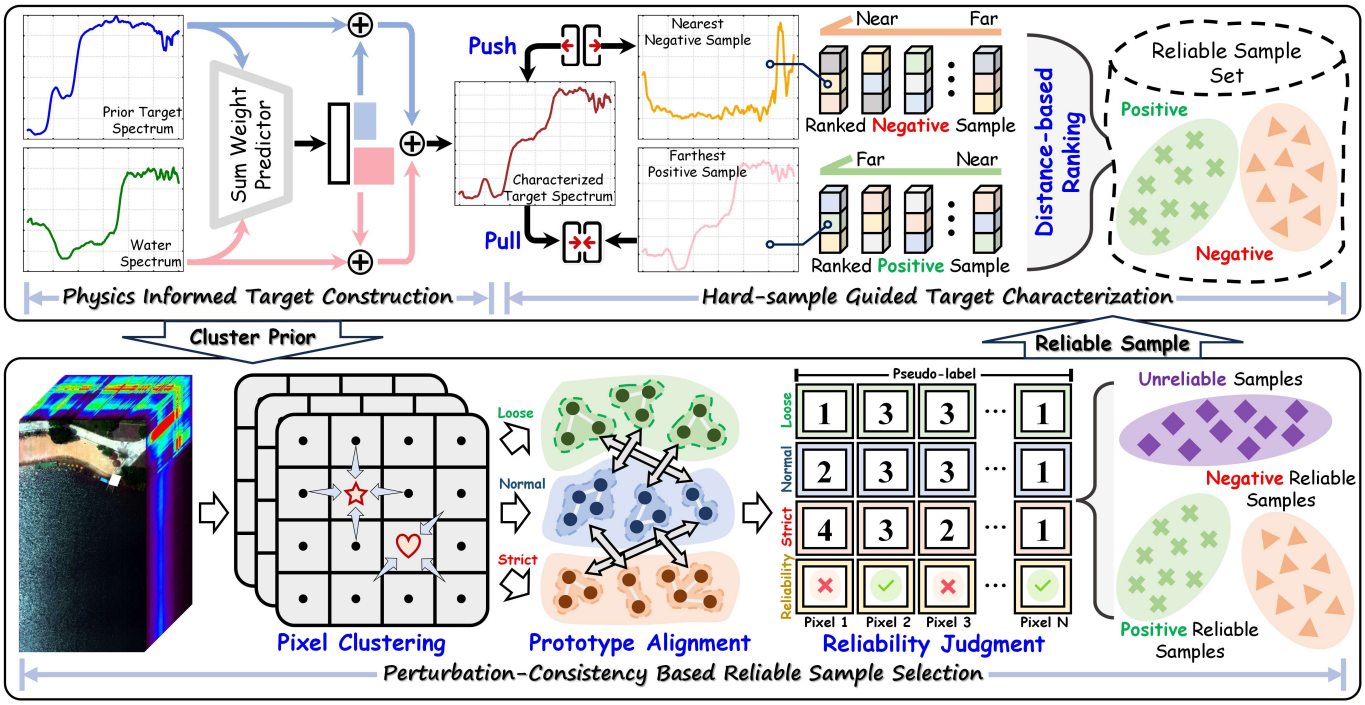



Fig. 4. Overall framework of PCL-HUTD. PCL-HUTD consists of three main components: a physics-guided target construction module, a contrastive characterization module, and a consistency-guided reliable sample selection strategy. The target construction module leverages the bathymetric model to provide physical interpretability, while the contrastive characterization module optimizes the characterized target spectrum for discriminability. The consistency-guided sample selection strategy uses unsupervised clustering to filter out unreliable samples based on their stability across different clustering criteria. Finally, a CL strategy is applied to progressively refine the target representation throughout training.

for learning basic discriminative features. As training progresses, the selection threshold is gradually relaxed to include more ambiguous samples, thereby implementing an *easy-to-hard learning schedule*. This approach enables the model to adapt to increasing sample uncertainty in a controlled manner while maintaining stability and robustness.

By integrating CL [35] into the iterative framework, the model achieves more stable convergence and improved generalization. In the end, the output generated by the target construction module is regarded as the characterized target spectrum, which can be directly used as the reference input for subsequent detection tasks.

## IV. Methodology

In this section, we present a PCL that enables robust underwater target characterization for HUTD tasks. The proposed framework integrates physical modeling with contrastive representation learning and unsupervised sample selection, forming a closed-loop structure that supports progressive refinement.

### A. Overall Framework

Based on the motivation discussed in Fig. 4, we design a unified framework named **PCL-HUTD** to address the major

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challenges of hyperspectral underwater target characterization. An overview of the proposed framework is shown in Fig. 4, which consists of four major components.

1) *Physics-Informed Target Construction:* A learnable module transforms the reference spectrum into an underwater target spectrum by simulating the underwater imaging process defined by the bathymetric model, ensuring physical interpretability and spectral plausibility.
2) *Hard-Sample-Guided Target Characterization (HSGTC):* The target construction module is optimized using a contrastive learning strategy that emphasizes hard examples. This strategy aims to maximize the spectral distance between the constructed target and background samples, while minimizing the distance to potential target samples.
3) *Perturbation-Consistency-Based Reliable Sample Selection:* To construct valid contrastive pairs without requiring annotations, we perform unsupervised clustering to infer the semantic category of each pixel. A perturbation-consistency-based mechanism is applied to select only the most stable and reliable samples.
4) *CL Strategy:* A progressive learning strategy is adopted to facilitate stable optimization, which gradually increases the difficulty of the learning task by relaxing sample selection criteria over training epochs.

The above-mentioned components are integrated into a closed-loop architecture, where the target characterization and sample selection modules reinforce each other. In Sections IV-B–IV-F, we elaborate on the design and implementation of each module and explain how they collaboratively contribute to effective underwater target characterization.

### B. Physics-Informed Target Construction

A straightforward approach for underwater target characterization is to directly learn the mapping from a reference spectrum to the corresponding underwater target spectrum using a deep neural network. However, in hyperspectral remote sensing, the limited availability of labeled data often restricts the effectiveness of fully data-driven methods [36], [37]. Moreover, the absence of physical constraints in such models can result in poor interpretability and limited generalization to unseen underwater scenarios [38]. To overcome these limitations, we propose a *physics-informed target construction module* that incorporates prior knowledge from the bathymetric imaging model to guide the learning process.

Let $\boldsymbol{I} \in \mathbb{R}^{H\times W\times C}$ represent the input hyperspectral underwater image, where $H$, $W$, and $C$ denote the height, width, and number of spectral bands, respectively. We estimate the water body spectrum $\boldsymbol{r}_w \in \mathbb{R}^C$ from the input image $\boldsymbol{I}$ using hyperspectral unmixing techniques, as described in [18] and [23]. According to (1), two fundamental properties derived from the underwater imaging model are utilized.

1) *Linear Combination:* The observed underwater spectrum is represented as a linear combination of the reference target spectrum $\boldsymbol{r}_t$ and water body spectrum $\boldsymbol{r}_w$.
2) *Sum-to-One Constraint:* The coefficients used in this combination are constrained to sum to one.

Based on these principles, the initial physically constructed underwater target spectrum $\hat{\boldsymbol{r}} \in \mathbb{R}^C$ is modeled as

$$\hat{\boldsymbol{r}} = \sigma(\alpha)\cdot \boldsymbol{r}_w + (1-\sigma(\alpha))\cdot \boldsymbol{r}_t \tag{2}$$

where $\alpha$ is a learnable scalar parameter, $\sigma(\cdot)$ denotes the sigmoid function used to restrict $\alpha$ to the interval $(0, 1)$.

Although the above-mentioned formulation encodes key physical constraints, the linear combination assumption in (2) may not fully capture the complexity of underwater target spectra, particularly in complex underwater environments [18]. To address this, we introduce a transformation network $\mathcal{F}_\theta(\cdot)$, parameterized by $\theta$, that maps $\hat{\boldsymbol{r}}$ into a nonlinear latent space. This transformation enhances the capacity of characterization representation to represent spectral distortions and nonlinear interactions caused by underwater conditions. It also enables better abstraction of task-relevant features and reduces redundancy inherent in hyperspectral data.

The transformation network $\mathcal{F}_\theta(\cdot)$ is implemented using $N$ 1-D convolutional layers with appropriate stride and padding settings. The network compresses the spectral dimension to $\lfloor C/32 \rfloor$, where $C$ is the number of original spectral bands. Each convolutional layer is followed by a ReLU activation function, and the final characterized target spectrum $\boldsymbol{r} \in \mathbb{R}^{\lfloor C/32 \rfloor}$ is computed as

$$\boldsymbol{r} = \mathcal{F}_\theta\left(\sigma(\alpha)\cdot \boldsymbol{r}_w + (1-\sigma(\alpha))\cdot \boldsymbol{r}_t\right). \tag{3}$$

In addition to its ability to capture nonlinear spectral variations, the transformation network acts as a feature extractor. It facilitates the learning of compact and semantically meaningful embeddings in the latent space, which serve as a more effective basis for distinguishing challenging samples in the subsequent hard-sample guided learning strategy.

### C. Hard-Sample-Guided Target Characterization

Following the motivation described in Section III-A, the objective of target characterization in this framework is to construct a spectrum $\boldsymbol{r}$ that captures the essential features of potential underwater targets while remaining clearly distinguishable from background spectra. Together with (3), this objective can be formulated as an optimization problem over the parameters $\theta$ and $\alpha$, aiming to simultaneously increase the dissimilarity between $\boldsymbol{r}$ and background spectra, and reduce the distance between $\boldsymbol{r}$ and potential target spectra.

Let $\boldsymbol{B} = \{\boldsymbol{b}_1, \boldsymbol{b}_2, \ldots, \boldsymbol{b}_{N_b}\}$ and $\boldsymbol{T} = \{\boldsymbol{t}_1, \boldsymbol{t}_2, \ldots, \boldsymbol{t}_{N_t}\}$ denote the sets of reliable background and target spectra, respectively, where $\boldsymbol{b}_i, \boldsymbol{t}_j \in \mathbb{R}^C$. We adopt a contrastive learning objective [39] to optimize the characterization parameters

$$\theta^*, \alpha^* = \arg\min_{\theta,\alpha} \mathcal{L}_{\mathrm{CL}}(\boldsymbol{r}, \boldsymbol{B}, \boldsymbol{T}) \tag{4}$$

with the contrastive loss defined as

$$\begin{aligned}\mathcal{L}_{\mathrm{CL}}(\boldsymbol{r}, \boldsymbol{B}, \boldsymbol{T}) = &-\frac{1}{N_b}\sum_{i=1}^{N_b}\log\left(1-\mathrm{sim}_{\mathrm{shift}}(\boldsymbol{r}, \mathcal{F}_\theta(\boldsymbol{b}_i))+\epsilon\right)\\ &-\frac{1}{N_t}\sum_{j=1}^{N_t}\log\left(\mathrm{sim}_{\mathrm{shift}}\left(\boldsymbol{r}, \mathcal{F}_\theta\left(\boldsymbol{t}_j\right)\right)+\epsilon\right)\end{aligned} \tag{5}$$

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where $\mathcal{F}_\theta(\cdot)$ projects spectra into the same latent space as $\boldsymbol{r}$, and $\epsilon$ is a small constant (e.g., $10^{-6}$) used to ensure numerical stability. The shifted cosine similarity is defined as

$$\mathrm{sim}_{\mathrm{shift}}(\boldsymbol{x},\boldsymbol{y}) = \frac{1+\frac{\boldsymbol{x}^T\cdot\boldsymbol{y}}{\|\boldsymbol{x}\|\|\boldsymbol{y}\|}}{2} \tag{6}$$

where $\|\cdot\|$ denotes the Euclidean norm.

Equation (5) considers all samples equally; however, this uniform treatment may lead to suboptimal performance due to shortcut learning in deep neural networks [40]. The model may focus on easy samples while ignoring more informative yet challenging ones, thus biasing the learned target spectrum. To address this limitation, we introduce a joint geometric—semantic hard sample mining mechanism [41], which selects the most informative examples based on their distance to $\boldsymbol{r}$. Specifically, we first compute the geometric distances between the characterized spectrum $\boldsymbol{r}$ and the latent representations of background and target samples

$$d_{b,i} = \|\boldsymbol{r} - \mathcal{F}_\theta(\boldsymbol{b}_i)\|, \quad d_{t,j} = \left\|\boldsymbol{r} - \mathcal{F}_\theta\left(\boldsymbol{t}_j\right)\right\| \tag{7}$$

where $d_{b,i}$ and $d_{t,j}$ denote the distances between $\boldsymbol{r}$ and the $i$th background and $j$th target samples, respectively.

In parallel, we quantify intraclass variability by measuring each sample's deviation from its corresponding class centroid

$$v_{b,i} = \left\|\mathcal{F}_\theta(\boldsymbol{b}_i) - \bar{\boldsymbol{b}}\right\|, \quad v_{t,j} = \left\|\mathcal{F}_\theta\left(\boldsymbol{t}_j\right) - \bar{\boldsymbol{t}}\right\| \tag{8}$$

where $\bar{\boldsymbol{b}}$ and $\bar{\boldsymbol{t}}$ represent the mean embeddings of the background and target classes, which are computed as

$$\bar{\boldsymbol{b}} = \frac{1}{N_b}\sum_{i=1}^{N_b}\mathcal{F}_\theta(\boldsymbol{b}_i), \quad \bar{\boldsymbol{t}} = \frac{1}{N_t}\sum_{j=1}^{N_t}\mathcal{F}_\theta\left(\boldsymbol{t}_j\right). \tag{9}$$

Both geometric distances and intraclass deviations are normalized to $[0,1]$ as

$$\begin{aligned}\hat{d}_{b,i} &= \frac{d_{b,i}-d_{b,\min}}{d_{b,\max}-d_{b,\min}+\epsilon}, \hat{d}_{t,j} = \frac{d_{t,j}-d_{t,\min}}{d_{t,\max}-d_{t,\min}+\epsilon}\\ \hat{v}_{b,i} &= \frac{v_{b,i}-v_{b,\min}}{v_{b,\max}-v_{b,\min}+\epsilon}, \hat{v}_{t,j} = \frac{v_{t,j}-v_{t,\min}}{v_{t,\max}-v_{t,\min}+\epsilon}\end{aligned} \tag{10}$$

where $d_{b,\min}$ and $d_{b,\max}$ ($d_{t,\min}$ and $d_{t,\max}$) denote the minimum and maximum distances of background (target) samples, $v_{b,\min}$ and $v_{b,\max}$ ($v_{t,\min}$ and $v_{t,\max}$) are the extrema of intraclass deviations, and $\epsilon = 10^{-6}$ is a small constant to avoid division by zero.

Then, their product defines the final weighted difficulty score

$$\tilde{d}_{b,i} = \hat{d}_{b,i}/\hat{v}_{b,i}, \quad \tilde{d}_{t,j} = \hat{d}_{t,j}\cdot\hat{v}_{t,j} \tag{11}$$

where $\hat{d}_{(\cdot)}$ and $\hat{v}_{(\cdot)}$ denote the normalized geometric distances and intraclass deviations, respectively.

Based on these weighted scores, the hardest samples are selected as

$$\boldsymbol{b}_{\mathrm{hard}} = \arg\min_i \tilde{d}_{b,i}, \quad \boldsymbol{t}_{\mathrm{hard}} = \arg\max_j \tilde{d}_{t,j} \tag{12}$$

where $\boldsymbol{b}_{\mathrm{hard}}$ corresponds to the background sample that is close to $\boldsymbol{r}$ while exhibiting high intraclass variability, and $\boldsymbol{t}_{\mathrm{hard}}$ denotes the target sample that is both farthest from $\boldsymbol{r}$ and semantically complex.

This semantic weighting strategy ensures that the model attends to the most informative hard cases, thereby avoiding the limitations of relying solely on geometric distance. We then define the hardest background and target samples as

$$\boldsymbol{b}_{\mathrm{hard}} = \arg\min_i d_{b,i}, \quad \boldsymbol{t}_{\mathrm{hard}} = \arg\max_j d_{t,j} \tag{13}$$

where $\boldsymbol{b}_{\mathrm{hard}}$ is the background spectrum most similar to $\boldsymbol{r}$, and $\boldsymbol{t}_{\mathrm{hard}}$ is the target spectrum most dissimilar to $\boldsymbol{r}$.

These hard examples are used to formulate a refined contrastive loss

$$\begin{aligned}\mathcal{L}_{\mathrm{CL}}^{\mathrm{hard}}(\boldsymbol{r},\boldsymbol{B},\boldsymbol{T}) = &-\log\left(1-\mathrm{sim}_{\mathrm{shift}}\left(\boldsymbol{r},\mathcal{F}_\theta\left(\boldsymbol{b}_{\mathrm{hard}}\right)\right)+\epsilon\right)\\ &-\log\left(\mathrm{sim}_{\mathrm{shift}}\left(\boldsymbol{r},\mathcal{F}_\theta\left(\boldsymbol{t}_{\mathrm{hard}}\right)\right)+\epsilon\right).\end{aligned} \tag{14}$$

Finally, the parameters $\theta$ and $\alpha$ are jointly optimized by minimizing the hard-sample-guided loss over $E_{\mathrm{char}}$ training epochs

$$\theta^*, \alpha^* = \arg\min_{\theta,\alpha}\mathcal{L}_{\mathrm{CL}}^{\mathrm{hard}}(\boldsymbol{r},\boldsymbol{B},\boldsymbol{T}). \tag{15}$$

### *D. Perturbation-Consistency-Based Reliable Sample Selection*

The reliable target and background samples are crucial for the HSGTC in Section IV-C, but they are unavailable in the HUTD task. To address this issue, we propose a perturbation-consistency-based reliable sample selection strategy to tailor unsupervised clustering for reliable sample selection, grounded in the theoretical analysis of Section III-B. As shown in Fig. 4, the proposed strategy comprises three steps: pixel clustering, prototype alignment, and reliability judgment.

*1) Pixel Clustering:* Given the pixel set $\{\boldsymbol{x}_i \in \mathbb{R}^C\}_{i=1}^N$ from the HSI $\boldsymbol{I}$, we apply a transformation $\mathcal{T}(\cdot)$ to project each pixel into a lower dimensional representation

$$\boldsymbol{x}_i' = \mathcal{T}(\boldsymbol{x}_i) = \begin{cases}\mathrm{PCA}(\boldsymbol{x}_i), & \text{if } E=1\\ \mathcal{F}_\theta(\boldsymbol{x}_i), & \text{if } E>1\end{cases} \tag{16}$$

where $\mathrm{PCA}(\cdot)$ represents principal component analysis, $\mathcal{F}_\theta(\cdot)$ is the learned transformation network, and $E$ is the current training epoch. In the first epoch, PCA prevents the use of randomly initialized features; in later epochs, the learned transformation $\mathcal{F}_\theta(\cdot)$ encodes characterization-relevant features, enhancing clustering performance.

To introduce perturbation, we vary the clustering radius $\epsilon_{\mathrm{normal}}$ of DBSCAN [42] algorithm by a margin $\Delta$, yielding three cluster sets: $\mathcal{C}_{\mathrm{strict}}$ with $\epsilon_{\mathrm{strict}} = \epsilon_0 - \Delta$, $\mathcal{C}_{\mathrm{normal}}$ with $\epsilon_{\mathrm{normal}} = \epsilon_0$, and $\mathcal{C}_{\mathrm{loose}}$ with $\epsilon_{\mathrm{loose}} = \epsilon_0 + \Delta$, where $\epsilon_0$ is the initial clustering radius. These clusterings yield sets of prototypes $\mathcal{P}_m = \{\boldsymbol{p}_m^k\}_{k=1}^{K_m}$ for $m \in \{\mathrm{strict}, \mathrm{normal}, \mathrm{loose}\}$, where $K_{\mathrm{strict}} \geq K_{\mathrm{normal}} \geq K_{\mathrm{loose}}$.

*2) Prototype Alignment:* Since the clusters from the three DBSCAN runs differ in size and semantics, identifying semantically consistent clusters across $\mathcal{C}_{\mathrm{normal}}$, $\mathcal{C}_{\mathrm{strict}}$, and $\mathcal{C}_{\mathrm{loose}}$ is nontrivial, which poses a challenge for subsequent reliability assessment. For this, we align their prototypes before reliability evaluation. Given that $\mathcal{C}_{\mathrm{loose}}$ contains the smallest number of clusters, it is selected as the reference, and both $\mathcal{C}_{\mathrm{normal}}$ and $\mathcal{C}_{\mathrm{strict}}$ are aligned to it.

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For $\mathcal{C}_{\text{normal}}$, we compute the cosine similarity between each pair of prototypes from $\mathcal{P}_{\text{loose}}$ and $\mathcal{P}_{\text{normal}}$, forming a similarity matrix $\boldsymbol{S}_{L\to N} \in \mathbb{R}^{K_{\text{loose}}\times K_{\text{normal}}}$. A higher value in $\boldsymbol{S}_{L\to N}(i,j)$ indicates greater similarity between $\boldsymbol{p}^i_{\text{loose}}$ and $\boldsymbol{p}^j_{\text{normal}}$. We then apply the Hungarian algorithm [43] to $\boldsymbol{S}_{L\to N}$ to obtain a one-to-one prototype mapping

$$\mathcal{M}_{L\to N} = \left\{\left(\boldsymbol{p}^1_{\text{loose}}, \boldsymbol{p}^l_{\text{normal}}\right), \left(\boldsymbol{p}^2_{\text{loose}}, \boldsymbol{p}^m_{\text{normal}}\right), \ldots, \left(\boldsymbol{p}^{K_{\text{loose}}}_{\text{loose}}, \boldsymbol{p}^n_{\text{normal}}\right)\right\} \tag{17}$$

where $\mathcal{M}_{L\to N}$ denotes the alignment mapping from $\mathcal{P}_{\text{loose}}$ to $\mathcal{P}_{\text{normal}}$, and each pair $(\boldsymbol{p}^i_{\text{loose}}, \boldsymbol{p}^j_{\text{normal}})$ indicates a matched prototype pair. Since $K_{\text{normal}} \geq K_{\text{loose}}$, some prototypes in $\mathcal{P}_{\text{normal}}$ may remain unmatched. Their corresponding clusters and pixels are excluded from further analysis. The same alignment procedure is applied between $\mathcal{P}_{\text{strict}}$ and $\mathcal{P}_{\text{loose}}$ to yield the mapping $\mathcal{M}_{L\to S}$

$$\mathcal{M}_{L\to S} = \left\{\left(\boldsymbol{p}^1_{\text{loose}}, \boldsymbol{p}^l_{\text{strict}}\right), \left(\boldsymbol{p}^2_{\text{loose}}, \boldsymbol{p}^m_{\text{strict}}\right), \ldots, \left(\boldsymbol{p}^{K_{\text{loose}}}_{\text{loose}}, \boldsymbol{p}^n_{\text{strict}}\right)\right\}. \tag{18}$$

*3) Reliability Judgment:* After prototype alignment, we evaluate the reliability of each pixel sample $\boldsymbol{x}_i$ based on the consistency of its cluster assignments across $\mathcal{C}_{\text{normal}}$, $\mathcal{C}_{\text{strict}}$, and $\mathcal{C}_{\text{loose}}$. For each pixel, we identify its assigned cluster index as

$$a^i_m = \arg\min_k \|\boldsymbol{x}'_i - \boldsymbol{p}^k_m\|, \quad m \in \{\text{normalstrict}, \text{loose}\} \tag{19}$$

where $a^i_m$ denotes the index of the closest prototype in $\mathcal{P}_m$.

To compare the cluster assignments across different sets, we use the prototype alignment mappings $\mathcal{M}_{L\to N}$ and $\mathcal{M}_{L\to S}$ to associate cluster indices in $\mathcal{C}_{\text{normal}}$ and $\mathcal{C}_{\text{strict}}$ with those in $\mathcal{C}_{\text{loose}}$. Let $\phi_{L\leftarrow N}(k)$ denote the index in $\mathcal{C}_{\text{loose}}$ that corresponds to the prototype $\boldsymbol{p}^k_{\text{normal}}$ under the mapping $\mathcal{M}_{L\to N}$, and similarly let $\phi_{L\leftarrow S}(k)$ denote the index in $\mathcal{C}_{\text{loose}}$ for $\boldsymbol{p}^k_{\text{strict}}$ under $\mathcal{M}_{L\to S}$. The reliability label $r_i$ is then defined by checking whether the cluster assignment of $\boldsymbol{x}_i$ remains consistent across all three cluster sets $\mathcal{C}_{\text{normal}}$, $\mathcal{C}_{\text{strict}}$, and $\mathcal{C}_{\text{loose}}$ after prototype alignment

$$r_i = \begin{cases} 1, & \text{if } \phi_{L\leftarrow N}\left(a^i_{\text{normal}}\right) = a^i_{\text{loose}} = \phi_{L\leftarrow S}\left(a^i_{\text{strict}}\right), \\ 0, & \text{otherwise} \end{cases} \tag{20}$$

where $r_i = 1$ indicates that the pixel $\boldsymbol{x}_i$ is consistently assigned to the same cluster across all three sets after prototype alignment, suggesting a reliable clustering decision. Conversely, $r_i = 0$ indicates inconsistency in cluster assignments, and thus unreliability.

Finally, we collect all pixels with $r_i = 1$ into the reliable sample set $\mathcal{R}$. To further categorize $\mathcal{R}$ into target and background samples, we identify the cluster prototype in $\mathcal{P}_{\text{loose}}$ closest to the reference spectrum $\boldsymbol{r}_t$ (or the updated $\boldsymbol{r}$ in subsequent iterations) and assign all corresponding samples to the reliable target set $\mathcal{R}_t$, while the remaining samples are assigned to the background set $\mathcal{R}_b$.

### E. CL Strategy

In the early stages of training, pixel clustering is performed in an unrefined latent space where feature representations lack sufficient structure. As a result, only a small portion of samples can be reliably identified. Incorporating too many samples at this point may introduce substantial label noise, thereby disrupting the optimization process and impeding convergence. As training progresses, the latent representations gradually become more discriminative, improving clustering quality and yielding a larger pool of reliable samples. In such cases, retaining a small number of samples may limit the diversity and representativeness of contrastive supervision.

To adaptively match the number of selected samples with their evolving reliability, we introduce a *CL strategy* that gradually increases the scale of sample selection for contrastive target characterization.

**Algorithm 1** Training Procedure of PCL-HUTD

**Input:** Hyperspectral image $\boldsymbol{I}$, reference spectrum $\boldsymbol{r}_t$, total training epochs $E_{\max}$, epochs for contrastive characterization $E_{\text{char}}$, base neighborhood threshold $\epsilon_0$, and initial perturbation radius $\Delta_0$.

**Output:** Optimized transformation parameters $\theta^*$ and $\alpha^*$ in $\mathcal{F}_\theta(\cdot)$.

Initialize transformation network $\mathcal{F}_\theta(\cdot)$ and scalar parameter $\alpha$ randomly.
**for** $E = 1$ **to** $E_{max}$ **do**
  **Step 1: Pixel Clustering**
  Apply the transformation $\mathcal{T}(\cdot)$ in Eq. (16) to each pixel $\boldsymbol{x}_i$ to obtain transformed representations $\boldsymbol{x}'_i$.
  Compute the current perturbation radius $\Delta_E$ using Eq. (21).
  Set the DBSCAN radii as $\epsilon_{\text{normal}} = \epsilon_0$, $\epsilon_{\text{strict}} = \epsilon_0 - \Delta_E$, and $\epsilon_{\text{loose}} = \epsilon_0 + \Delta_E$.
  Perform DBSCAN clustering with the three radii to obtain cluster sets $\mathcal{C}_{\text{normal}}$, $\mathcal{C}_{\text{strict}}$, and $\mathcal{C}_{\text{loose}}$.
  **Step 2: Prototype Alignment**
  Use the Hungarian algorithm to align prototypes of $\mathcal{C}_{\text{normal}}$ and $\mathcal{C}_{\text{strict}}$ to those of $\mathcal{C}_{\text{loose}}$, resulting in mappings $\mathcal{M}_{L\to N}$ and $\mathcal{M}_{L\to S}$.
  **Step 3: Reliability Estimation**
  Assign each transformed pixel $\boldsymbol{x}'_i$ to its nearest prototype in each cluster set based on Eq. (19).
  Evaluate pixel-level reliability based on consistency across cluster assignments via Eq. (20).
  Construct the reliable sample set $\mathcal{R}$ and divide it into target $\mathcal{R}_t$ and background $\mathcal{R}_b$ subsets.
  **Step 4: Contrastive Optimization**
  Estimate the characterized spectrum $\hat{\boldsymbol{r}}_t$ using the physics-informed formulation in Eq. (3).
  Select the hardest positive and negative samples from $\mathcal{R}_t$ and $\mathcal{R}_b$ based on Eq. (13).
  Update the parameters $\theta$ and $\alpha$ by minimizing the hard-aware contrastive loss $\mathcal{L}_{\text{contrast}}$ in Eq. (15) for $E_{\text{char}}$ epochs.
**end**

As discussed in Section IV-D, the cardinality of the reliable sample set $\mathcal{R}$ is implicitly governed by the perturbation radius $\Delta$ in the clustering process. A larger $\Delta$ leads to greater inconsistencies among the cluster sets $\mathcal{C}_{\text{normal}}$, $\mathcal{C}_{\text{strict}}$, and $\mathcal{C}_{\text{loose}}$,

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TABLE I
DETAILED SPECIFICATIONS OF THE ATR2-HUTD BENCHMARK

| Dataset | | Wavelength | Spectral Resolution | Image Size | Spatial Resolution | Target Type | Target Depth |
|---|---|---|---|---|---|---|---|
| **Lake** | Scene1 | 400-1000nm | 2.2nm | 2304×640 pxiels | 5.55 cm | Black Metal Plate | 1.69m, 2.74m |
| | Scene2 | | | | | Blue Metal Plate | 0.91m, 1.28m |
| **River** | Scene1 | | | 3536×640 pxiels | 4.63 cm | Black Plastic Plate | 1.97m, 1.89m |
| | Scene2 | | | | | Black Metal Plate | 1.15m, 2.08m |
| **Sea** | Scene1 | | | 3171×640 pxiels | 2.78 cm | Black Wooden Board | 0.64m, 1.48m |
| | Scene2 | | | | | Yellow Wooden Board | 1.35m |

resulting in fewer samples passing the consistency filter. Building on this observation, we design a curriculum schedule that gradually decreases the value of $\Delta$ over training epochs, thereby expanding the reliable sample set in a controlled and progressive manner. Specifically, the perturbation radius $\Delta$ at epoch $E$ is defined as

$$\Delta_E = \Delta_0 \cdot \left(1 - \left(\frac{E}{E_{\max}}\right)^2\right) \tag{21}$$

where $\Delta_0$ is the initial perturbation magnitude and $E_{\max}$ denotes the total number of training epochs. This quadratic decay schedule imposes a slow decrease in the early phase to ensure training stability, while it accelerates the reduction in later stages to incorporate harder samples as the model gains sufficient capacity. The proposed curriculum strategy effectively aligns the model's learning capability with the evolving reliability of contrastive supervision. The complete training procedure of the proposed PCL-HUTD framework is summarized in Algorithm 1.

### F. Target Detection

Upon obtaining the optimized parameters $\theta^*$ and $\alpha^*$, the final characterized underwater target spectrum $\boldsymbol{r}$ is derived via (3). This spectrum is then employed to perform pixelwise detection over the hyperspectral underwater image $\boldsymbol{I}$, yielding a detection map $\boldsymbol{M} \in \mathbb{R}^{H\times W}$. Each element $\boldsymbol{M}_{i,j}$ indicates the spectral similarity between the pixel located at $(i, j)$ in $\boldsymbol{I}$ and the characterized spectrum $\boldsymbol{r}$

$$\boldsymbol{M}_{i,j} = \mathcal{D}\left(\mathcal{F}_\theta\left(\boldsymbol{I}_{i,j}\right), \boldsymbol{r}\right) \tag{22}$$

where $\boldsymbol{I}_{i,j} \in \mathbb{R}^C$ denotes the spectral vector of the pixel at location $(i, j)$. The similarity function $\mathcal{D}(\cdot,\cdot)$ quantifies the closeness between two spectral representations. It can be instantiated using any conventional HTD metric, such as spectral MF [45] and spectral angle mapper (SAM) [44]. This modular formulation enables flexible integration with existing detection algorithms while benefiting from the characterized target spectrum $\boldsymbol{r}$ produced by the proposed framework.

## V. EXPERIMENTS AND ANALYSIS

To validate the effectiveness and robustness of the proposed PCL-HUTD framework, this section presents a series of comprehensive experiments conducted on six real-world hyperspectral underwater datasets. First, Section V-A introduces the benchmark datasets used for evaluation. Section V-B then outlines the experimental settings, including evaluation metrics, implementation details, and baseline methods. Quantitative and qualitative comparisons with SOTA approaches are reported in Section V-C. Ablation studies analyzing the individual contribution of each core component are presented in Section V-D. Finally, Section V-E investigates the sensitivity of the proposed method to key hyperparameters, offering further insight into its design and stability.

### A. Datasets

The proposed method is evaluated on the ATR2-HUTD benchmark,[1] a large-scale UAV-borne hyperspectral dataset encompassing six real-world underwater scenes: Lake Scene1, Lake Scene2, River Scene1, River Scene2, Sea Scene1, and Sea Scene2. The lake scenes were captured over Qianlu Lake (Liuyang City), the river scenes over the Xiangjiang River (Changsha City), and the sea scenes near the coastal waters of Sanya City, China. All data were collected using a Nano-Hyperspec sensor (Headwall Photonics) mounted on a DJI Matrice 300 RTK UAV platform. This benchmark provides a diverse set of underwater environments with varying water types, enabling a comprehensive evaluation of model robustness under different optical conditions, including turbidity, salinity, and spectral attenuation. In addition, the dataset includes multiple types of submerged targets with distinct visual and spectral characteristics, such as Blue Metal Plate, Black Plastic Plate, and Yellow Wooden Board, which allow for detailed assessment of material- and color-based detection generalization. Target depths range from 0.64 to 2.74 m, facilitating evaluation of the method's adaptability under different levels of submersion and spectral distortion. The detailed information of the ATR2-HUTD benchmark are summarized in Table I.

### B. Experimental Setup

*1) Evaluation Metrics:* To comprehensively evaluate the proposed method, we employ a set of complementary performance metrics, including the detection map, receiver operating

[1]The benchmark is available at https://github.com/qjh1996/HUTD

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characteristic (ROC) curves, area under the ROC curve (AUC) scores, and TBS maps. These metrics jointly reflect the accuracy, robustness, and feature discriminability of HUTD models under varying scene complexities.

*a) Detection map visualization:* We first provide intuitive detection maps to visualize the pixelwise prediction responses of different methods. These maps highlight the spatial precision of target localization and the suppression of background activations. Qualitative comparisons across multiple scenes offer insights into visual fidelity and false-alarm characteristics.

*b) ROC curve analysis (3-D and 2-D):* We adopt both 3-D and 2-D ROC curve representations to evaluate detection reliability under varying decision thresholds $\tau$. The *3-D ROC curve* jointly considers the TPR, FPR, and threshold $\tau$, providing a comprehensive surface-level understanding of method performance. The *2-D ROC curves* are analyzed along three planes: 1) FPR versus TPR for standard discriminability; 2) threshold versus TPR for sensitivity of target detection over thresholds; and 3) threshold versus FPR for stability of background suppression.

*c) AUC-based quantitative metrics:* The scalar AUC values are computed by integrating over the corresponding ROC curves. Specifically, $\mathrm{AUC}_{(D,F)}$ evaluates the discriminative ability between target and background responses. $\mathrm{AUC}_{(F,\tau)}$ measures the stability of background prediction across thresholds, and lower is better. $\mathrm{AUC}_{(D,\tau)}$ reflects the robustness of target recall under varying confidence levels. To provide an overall performance index, we define a composite score

$$\mathrm{AUC}_{\mathrm{OA}} = \mathrm{AUC}_{(D,F)} + \mathrm{AUC}_{(D,\tau)} - \mathrm{AUC}_{(F,\tau)} \tag{23}$$

which jointly considers target sensitivity, background suppression, and threshold robustness.

Furthermore, we propose a signal-to-noise-oriented robustness metric

$$\mathrm{AUC}_{\mathrm{SNPR}} = \frac{\mathrm{AUC}_{(D,F)}}{\mathrm{AUC}_{(F,\tau)}} \tag{24}$$

which highlights the detector's ability to maintain high discriminability with minimal false alarms.

*d) TBS map:* To further analyze the feature-level discriminative capacity of each method, we employ the TBS map, which visualizes the distribution of detection responses for target and background pixels via statistical boxplots. Green and blue boxes denote the response distributions for target and background samples, respectively. Greater separation between the two indicates stronger target-background contrast in the learned feature space.

*2) Comparison Methods:* We compare the proposed method with several SOTA approaches, including two traditional HTD methods (MF [45] and SAM [44]), two unsupervised HTD methods (UADNet [46] and PST [47]), three recent deep learning-based HTD methods (IEEPST [48], MCLT [49], and PEPTE [50]), and four representative SOTA HUTD methods (UTD-Net [23], TUTDF [51], NUUTD [18], and HUTDNet [19]). Furthermore, since the proposed PCL-HUTD adopts a characterization-based strategy, we also include two representative characterization-based methods, MI [27] and MI-SP [29], for comparison. As characterization-based methods require a downstream detector to produce final predictions, we adopt a unified evaluation protocol by applying the same detection module, i.e., the MF method [45], to all such approaches, including PCL-HUTD. The MF algorithm is a widely used baseline in HTD, which allows us to fairly evaluate the contribution of different characterization modules independent of the detection mechanism.

TABLE II

SCENE-ADAPTIVE PARAMETER SETTINGS FOR $\epsilon_0$ (BASE NEIGHBORHOOD THRESHOLD) AND $\Delta_0$ (INITIAL PERTURBATION RADIUS)

| Scene | $\epsilon_0$ | $\Delta_0$ |
|---|---|---|
| Lake Scene1 | 0.418 | 0.169 |
| Lake Scene2 | 0.433 | 0.184 |
| River Scene1 | 0.548 | 0.304 |
| River Scene2 | 0.556 | 0.313 |
| Sea Scene1 | 0.474 | 0.227 |
| Sea Scene2 | 0.479 | 0.233 |

*3) Implementation Details:* In the physics-informed construction module, following [18], the VCA algorithm [52] is applied to unmix the input HSI and identify the water spectrum $\boldsymbol{r}_w(\lambda)$ as the endmember with the highest abundance. The transformation network $\mathcal{F}_\theta$ is implemented as a cascade of five 1-D convolutional layers, each with a stride of 2 to progressively reduce the spectral sequence length. The first two layers adopt a kernel size of 3 and padding of 1 to extract fine-grained spectral features, while the remaining three use a kernel size of 5 and padding of 2 to capture broader contextual information. In the hard-sample guided characterization module, the number of training epochs $E_{\mathrm{char}}$ is set to 100, with the AdamW optimizer and an initial learning rate of $5 \times 10^{-4}$. For the reliable sample selection module, the output dimension of the PCA transformation is set to $\lfloor C/32 \rfloor$, where $C$ denotes the number of spectral bands, in order to match the output dimension of $\mathcal{F}_\theta$. Following a scene-adaptive strategy described in Section V of the Supplementary Material, we set the base neighborhood thresholds $\epsilon_0$ and initial perturbation radii $\Delta_0$ for each scene of the ATR2-HUTD benchmark in Table II. The corresponding influence and rationale behind these hyperparameter settings are further discussed in Section V-E. All experiments are conducted on a workstation equipped with seven NVIDIA A6000 GPUs, an AMD Ryzen 5995WX processor, 128 GB of RAM, and running Ubuntu 22.04.

### C. Performance Comparison With SOTA Methods

To validate the effectiveness of the proposed PCL-HUTD framework, we conduct comprehensive experiments on the ATR2-HUTD benchmark and compare it with a diverse set of SOTA methods. Performance is assessed from four complementary perspectives: *visual* (detection maps), *qualitative* (ROC curves), *quantitative* (AUC metrics), and *separability* (TBS maps).

*1) Detection Maps:* Due to the content length limitation, only one representative detection map for each scene category

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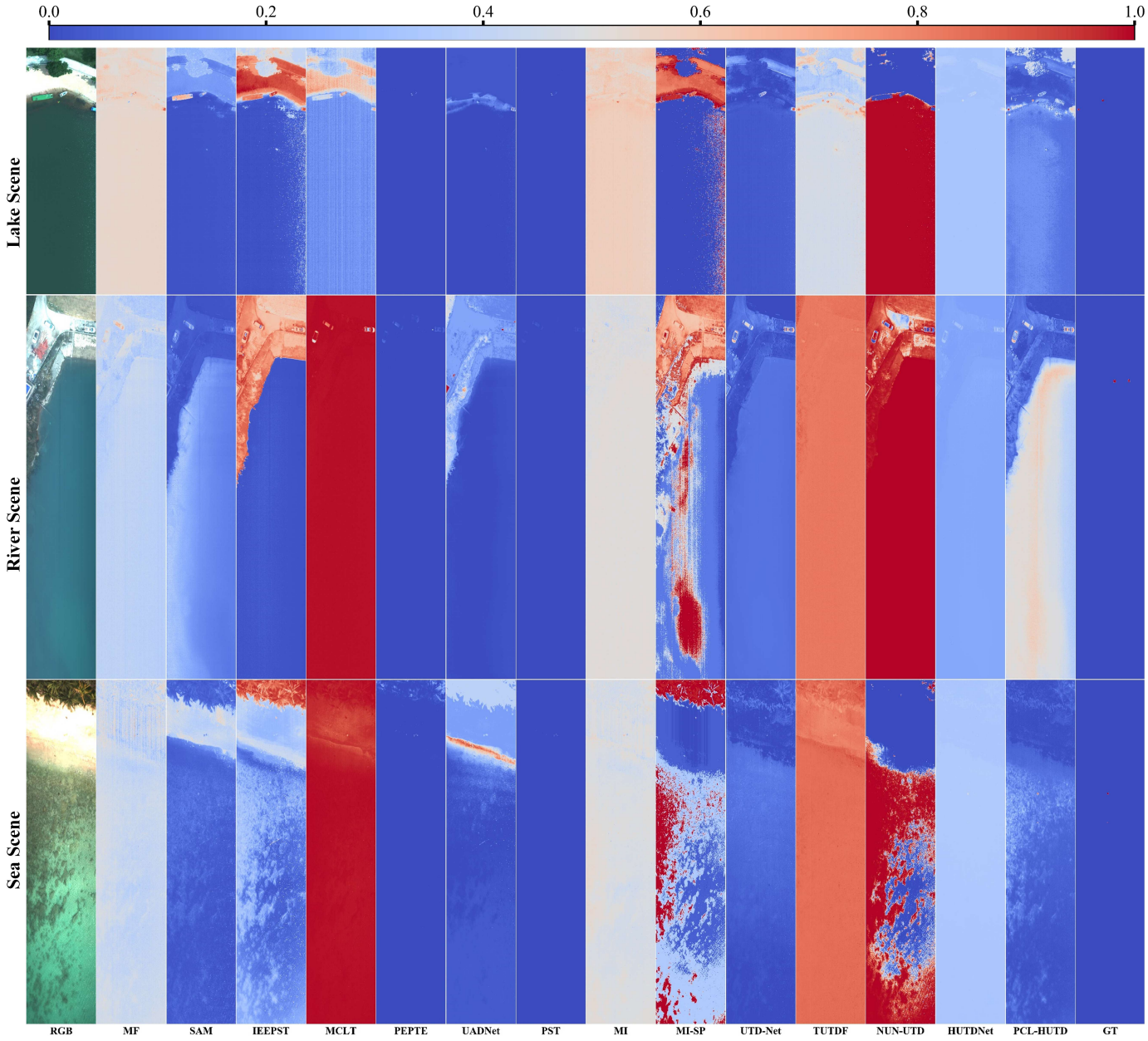



Fig. 5. Detection maps of the proposed PCL-HUTD method and comparison methods on one representative scene from each subdataset. The first row shows the detection maps of the Lake subdataset, the second row shows those of the River subdataset, and the third row shows those of the Sea subdataset.

(*Lake*, *River*, and *Sea*) is presented in Fig. 5. Traditional detectors, such as MF and SAM, generate overly smoothed responses that often fail to differentiate targets from surrounding water, primarily due to low spectral contrast. In terms of the unsupervised HTD methods, UADNet and PST exhibit conservative behavior with clean backgrounds and a few false alarms, but at the cost of weak target responses. Deep learning-based methods, including IEEPST and MCLT, produce more prominent activation regions. However, these responses are frequently accompanied by false positives, especially in shoreline and vegetative regions, reflecting limited robustness against spectral interference and background complexity. Among the SOTA HUTD methods, UTD-Net, TUTDF, and NUUTD demonstrate improved target localization. Nevertheless, their results still exhibit fragmented predictions or spurious activations in textured background areas, particularly in the challenging *Sea* scene. These methods often struggle to maintain spatial compactness under complex environmental variations. For characterization-based methods, both MI and MI-SP are evaluated under the same detection protocol using MF to ensure a fair comparison. MI often produces sparse detections with significant target omissions, while MI-SP activates extensive background regions, suggesting difficulty in preserving fine-grained TBS when relying on a common detector. While *PEPTE* and *HUTDNet* benefit from physics-informed modeling, their performance remains constrained by fragmented targets and oversmoothed activations in complex underwater environments.

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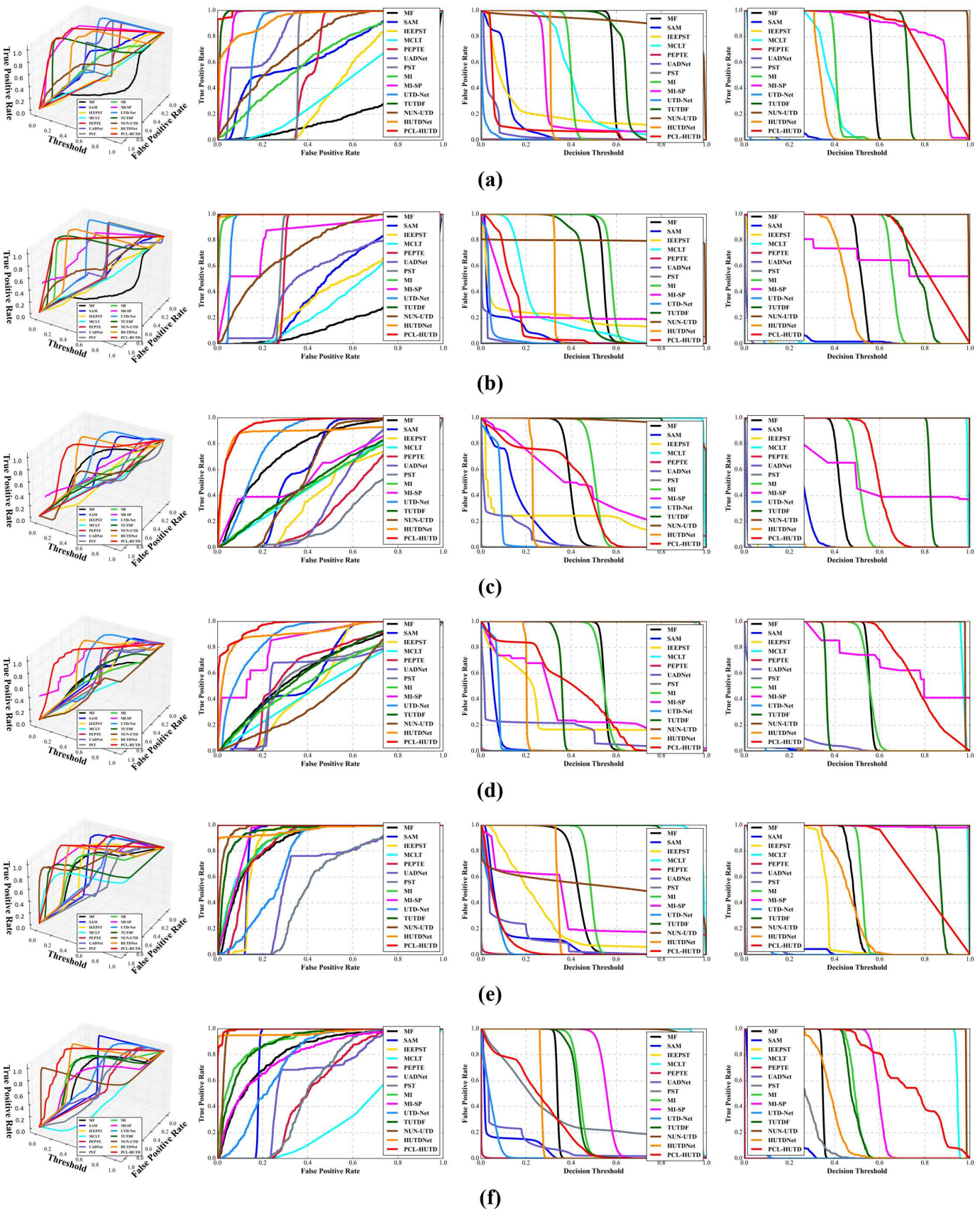



Fig. 6. ROC maps of different scenes. (a) Lake Subdataset Scene 1. (b) Lake Subdataset Scene 2. (c) River Subdataset Scene 1. (d) River Subdataset Scene 2. (e) Sea Subdataset Scene 1. (f) Sea Subdataset Scene 2. From (left to right) 3-D ROC surface, 2-D ROC curve (FPR versus TPR), 2-D curve (Threshold versus TPR), and 2-D curve (Threshold versus FPR).

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In contrast, the proposed PCL-HUTD method consistently yields clean, compact, and high-fidelity detection maps across all scenes. It delineates target regions with sharp boundaries and maintains low background activation. In the particularly complex *River* scene, PCL-HUTD successfully highlights small-scale targets located near structural edges and effectively suppresses responses from shadowed water regions. Such regions frequently cause failures in other methods. These qualitative comparisons demonstrate the superiority of the proposed contrastive characterization strategy. Even when combined with a simple MF detector, PCL-HUTD significantly enhances the semantic separability and robustness of target representations. It achieves reliable and interpretable performance across diverse and challenging underwater environments.

*2) ROC Curves:* To further evaluate quantitative detection performance, we present a comprehensive set of ROC-based evaluation maps for each scene in Fig. 6, including 3-D ROC surfaces and multiple 2-D projections. Each row corresponds to a specific subscene.

*Three-D ROC Surfaces* (*first column*) provide a holistic view of model behavior by jointly considering threshold values, FPR, and TPR. The proposed PCL-HUTD (in red) consistently exhibits smooth and elevated ROC surfaces across all scenes, suggesting stable sensitivity across a wide range of decision thresholds. Besides, it also shows that UADNet and PST maintain relatively flat surfaces with limited elevation, reflecting their conservative detection strategy. In contrast, methods, such as NUUTD (in saddlebrown) and MI-SP (in magenta), show abrupt gradients or irregular peaks, indicating limited adaptability to threshold variations and a higher risk of overfitting.

*Two-Dimensional ROC (FPR versus TPR)* projections (*second column*) offer a conventional perspective on model discriminability. The PCL-HUTD consistently achieves favorable tradeoffs between sensitivity and specificity, outperforming traditional methods, like MF (in black) and SAM (in blue), and often surpassing deep learning-based methods, such as IEEPST (in gold) and MCLT (in cyan). While NUUTD occasionally reaches higher TPRs, these gains are often accompanied by a sharp increase in FPR, as evidenced by its lower performance in threshold-related plots. Besides, *PEPTE* and *HUTDNet* reach competitive performance in lake scenes, but their curves flatten in the river and sea cases, with TPR gains accompanied by a noticeable rise in FPR.

*Two-Dimensional ROC (Threshold versus TPR)* plots (*third column*) analyze model recall characteristics across varying confidence thresholds. Although NUUTD often attains high TPRs throughout the threshold range, this behavior reflects an aggressive detection strategy that tends to label a large portion of the image as target. Moreover, it also confirms that the recall of UADNet and PST remains low across nearly all thresholds, which is consistent with their detection maps that miss small or weak targets. This results in inflated recall scores at the expense of precision. Such tendencies are visually confirmed by detection maps, which reveal substantial overactivation and imprecise localization for this method.

*Two-Dimensional ROC (Threshold versus FPR)* plots (*fourth column*) illustrate how the background false alarm rate varies as the threshold is lowered. Methods, such as MF and MI, exhibit relatively low FPRs, but this outcome is mainly due to their conservative decision boundaries, which often fail to activate on true targets. In addition, both *PEPTE* and *HUTDNet* experience difficulty in suppressing background activations when thresholds are lowered, especially in the more complex river and sea scenes. In contrast, PCL-HUTD maintains a balanced profile, achieving moderate FPR while preserving high recall, thereby avoiding the extremes of overdetection or underdetection. Together, these ROC-based evaluations confirm the conclusions drawn from detection map comparisons. Models that perform well under a single metric may exhibit significant weaknesses under more comprehensive evaluations. By contrast, PCL-HUTD consistently maintains robust performance across all ROC dimensions, underscoring its practical reliability and superior generalization in diverse real-world HUTD scenarios.

*3) AUC Scores:* To complement the ROC visualizations, we report a set of AUC-based quantitative indicators in Table III, which summarize detection performance across six subscenes. Specifically, we include: 1) $\mathrm{AUC}_{(D,F)}$, representing the area under the standard ROC curve (FPR versus TPR); 2) $\mathrm{AUC}_{(F,\tau)}$, quantifying the false positive rate as a function of threshold; 3) $\mathrm{AUC}_{(D,\tau)}$, indicating the true positive rate across threshold values; 4) $\mathrm{AUC}_{(\mathrm{OA})}$, representing the overall accuracy; and 5) $\mathrm{AUC}_{(\mathrm{SNPR})}$, a signal-to-noise-oriented robustness metric that balances target sensitivity and background suppression.

Across all evaluated scenes, the proposed PCL-HUTD achieves the highest $\mathrm{AUC}_{(D,F)}$ in five out of six cases, demonstrating consistent superiority in the FPR-TPR domain. While methods, such as NUUTD occasionally attain slightly higher $\mathrm{AUC}_{(D,\tau)}$ values, their overall performance is undermined by significantly larger $\mathrm{AUC}_{(F,\tau)}$ scores. This observation indicates that their high recall is achieved by indiscriminately classifying a substantial proportion of pixels as targets, as further evidenced by their overly activated detection maps. This tradeoff is clearly reflected in both the $\mathrm{AUC}_{\mathrm{OA}}$ and $\mathrm{AUC}_{\mathrm{SNPR}}$ scores. Unsupervised clustering-based baselines such as UADNet and PST present the opposite tradeoff: they suppress false alarms to extremely low levels (e.g., $\mathrm{AUC}_{(F,\tau)}$ close to zero in the *River* and *Sea* scenes), but at the cost of vanishing recall, with $\mathrm{AUC}_{(D,\tau)}$ values near zero and very limited overall accuracy. These conservative behaviors highlight their inability to capture weak or small targets in complex underwater environments. At the same time, physics-informed methods, such as PEPTE and HUTDNet, achieve competitive results compared with traditional detectors, but their improvements remain limited under complex underwater conditions. The PCL-HUTD consistently ranks highest in $\mathrm{AUC}_{\mathrm{OA}}$ across all six scenes, highlighting its ability to maintain a favorable balance between sensitivity and specificity under varying threshold conditions. For example, in the *Sea Scene 1*, PCL-HUTD achieves an $\mathrm{AUC}_{\mathrm{OA}}$ of 1.746 and an exceptional $\mathrm{AUC}_{\mathrm{SNPR}}$ of 24.337. This value is more than nine times greater than that of the next best-performing method, underscoring its robustness in minimizing both missed detections and false alarms. It is

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TABLE III

QUANTITATIVE COMPARISON RESULTS. THE BEST AND SECOND BEST RESULTS ARE IN **BOLD** AND WITH UNDERLINE

| Method | ATR2-HUTD-Lake Scene1 | | | | | ATR2-HUTD-Lake Scene2 | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | $AUC_{(D,F)}\uparrow$ | $AUC_{(F,\tau)}\downarrow$ | $AUC_{(D,\tau)}\uparrow$ | $AUC_{OA}\uparrow$ | $AUC_{SNPR}\uparrow$ | $AUC_{(D,F)}\uparrow$ | $AUC_{(F,\tau)}\downarrow$ | $AUC_{(D,\tau)}\uparrow$ | $AUC_{OA}\uparrow$ | $AUC_{SNPR}\uparrow$ |
| MF | 0.201 | 0.588 | 0.573 | 0.187 | 0.976 | 0.190 | 0.545 | 0.519 | 0.164 | 0.953 |
| SAM | 0.670 | 0.119 | 0.119 | 0.670 | 0.994 | 0.480 | 0.069 | 0.079 | 0.491 | 1.155 |
| IEEPST | 0.424 | 0.204 | 0.075 | 0.295 | 0.369 | 0.417 | 0.187 | 0.036 | 0.266 | 0.193 |
| MCLT | 0.401 | 0.422 | 0.377 | 0.357 | 0.894 | 0.365 | 0.243 | 0.173 | 0.296 | 0.715 |
| PEPTE | 0.592 | 0.005 | 0.001 | 0.589 | 0.211 | 0.706 | **0.003** | 0.001 | 0.704 | 0.377 |
| UADNet | 0.852 | 0.049 | 0.087 | 0.890 | 1.780 | 0.532 | 0.014 | 0.009 | 0.527 | 0.664 |
| PST | 0.644 | **0.002** | 0.001 | 0.643 | 0.466 | 0.723 | 0.004 | 0.001 | 0.720 | 0.279 |
| MI | 0.643 | 0.400 | 0.411 | 0.654 | 1.027 | 0.992 | 0.579 | 0.659 | 1.073 | 1.140 |
| MI-SP | 0.962 | 0.329 | 0.847 | 1.480 | 2.577 | 0.839 | 0.236 | 0.697 | 1.300 | 2.959 |
| UTD-Net | 0.846 | 0.013 | 0.019 | 0.853 | 1.492 | 0.944 | 0.041 | 0.073 | 0.976 | 1.771 |
| TUTDF | 0.990 | 0.634 | 0.726 | 1.081 | 1.145 | 0.998 | 0.461 | 0.768 | 1.306 | 1.667 |
| NUN-UTD | 0.758 | 0.913 | **0.994** | 0.838 | 1.088 | 0.765 | 0.792 | **0.995** | 0.968 | 1.256 |
| HUTDNet | 0.939 | 0.311 | 0.340 | 0.968 | 1.094 | 0.996 | 0.329 | 0.447 | 1.115 | 1.360 |
| PCL-HUTD | **0.992** | 0.092 | 0.861 | **1.761** | **9.346** | **0.999** | 0.112 | 0.876 | **1.763** | **7.823** |

| Method | ATR2-HUTD-River Scene1 | | | | | ATR2-HUTD-River Scene2 | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | $AUC_{(D,F)}\uparrow$ | $AUC_{(F,\tau)}\downarrow$ | $AUC_{(D,\tau)}\uparrow$ | $AUC_{OA}\uparrow$ | $AUC_{SNPR}\uparrow$ | $AUC_{(D,F)}\uparrow$ | $AUC_{(F,\tau)}\downarrow$ | $AUC_{(D,\tau)}\uparrow$ | $AUC_{OA}\uparrow$ | $AUC_{SNPR}\uparrow$ |
| MF | 0.780 | 0.396 | 0.423 | 0.807 | 1.067 | 0.659 | 0.548 | 0.556 | 0.667 | 1.014 |
| SAM | 0.657 | 0.156 | 0.210 | 0.711 | 1.349 | 0.656 | 0.061 | 0.074 | 0.669 | 1.210 |
| IEEPST | 0.455 | 0.203 | 0.033 | 0.286 | 0.163 | 0.594 | 0.274 | 0.236 | 0.556 | 0.862 |
| MCLT | 0.550 | 0.989 | 0.990 | 0.552 | 1.001 | 0.531 | 0.970 | 0.971 | 0.533 | 1.002 |
| PEPTE | 0.381 | **0.002** | 0.001 | 0.380 | 0.643 | 0.656 | **0.002** | 0.001 | 0.654 | 0.439 |
| UADNet | 0.442 | 0.064 | 0.010 | 0.388 | 0.156 | 0.605 | 0.134 | 0.098 | 0.569 | 0.729 |
| PST | 0.280 | 0.003 | 0.001 | 0.278 | 0.340 | 0.579 | 0.002 | 0.001 | 0.578 | 0.431 |
| MI | 0.554 | 0.505 | 0.512 | 0.560 | 1.013 | 0.597 | 0.545 | 0.555 | 0.607 | 1.019 |
| MI-SP | 0.633 | 0.454 | 0.606 | 0.785 | 1.334 | 0.849 | 0.348 | 0.744 | 1.245 | 2.140 |
| UTD-Net | 0.843 | 0.080 | 0.096 | 0.860 | 1.208 | 0.889 | 0.075 | 0.088 | 0.903 | 1.176 |
| TUTDF | 0.568 | 0.822 | 0.824 | 0.570 | 1.003 | 0.659 | 0.356 | 0.363 | 0.667 | 1.022 |
| NUN-UTD | 0.632 | 0.968 | **0.999** | 0.663 | 1.032 | 0.503 | 0.977 | **0.979** | 0.505 | 1.002 |
| HUTDNet | 0.900 | 0.229 | 0.246 | 0.917 | 1.073 | 0.920 | 0.203 | 0.217 | 0.934 | 1.072 |
| PCL-HUTD | **0.954** | 0.383 | 0.720 | **1.291** | **1.879** | **0.976** | 0.357 | 0.789 | **1.408** | **2.210** |

| Method | ATR2-HUTD-Sea Scene1 | | | | | ATR2-HUTD-Sea Scene2 | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | $AUC_{(D,F)}\uparrow$ | $AUC_{(F,\tau)}\downarrow$ | $AUC_{(D,\tau)}\uparrow$ | $AUC_{OA}\uparrow$ | $AUC_{SNPR}\uparrow$ | $AUC_{(D,F)}\uparrow$ | $AUC_{(F,\tau)}\downarrow$ | $AUC_{(D,\tau)}\uparrow$ | $AUC_{OA}\uparrow$ | $AUC_{SNPR}\uparrow$ |
| MF | 0.872 | 0.430 | 0.484 | 0.926 | 1.125 | 0.828 | 0.339 | 0.348 | 0.837 | 1.027 |
| SAM | 0.866 | 0.101 | 0.077 | 0.842 | 0.757 | 0.819 | 0.057 | 0.055 | 0.817 | 0.961 |
| IEEPST | 0.850 | 0.252 | 0.363 | 0.961 | 1.440 | 0.580 | 0.326 | 0.269 | 0.523 | 0.825 |
| MCLT | 0.895 | 0.980 | 0.994 | 0.909 | 1.014 | 0.317 | 0.953 | 0.945 | 0.309 | 0.991 |
| PEPTE | 0.861 | 0.004 | 0.002 | 0.860 | 0.707 | 0.567 | **0.003** | 0.001 | 0.566 | 0.364 |
| UADNet | 0.624 | 0.098 | 0.073 | 0.599 | 0.743 | 0.625 | 0.092 | 0.054 | 0.586 | 0.581 |
| PST | 0.550 | **0.002** | 0.001 | 0.549 | 0.534 | 0.580 | 0.326 | 0.269 | 0.523 | 0.825 |
| MI | 0.891 | 0.487 | 0.523 | 0.927 | 1.074 | 0.915 | 0.439 | 0.494 | 0.970 | 1.126 |
| MI-SP | 0.922 | 0.363 | 0.993 | 1.552 | 2.736 | 0.813 | 0.557 | 0.595 | 0.851 | 1.068 |
| UTD-Net | 0.762 | 0.050 | 0.083 | 0.796 | 1.678 | 0.774 | 0.043 | 0.070 | 0.801 | 1.631 |
| TUTDF | 0.952 | 0.841 | 0.872 | 0.984 | 1.037 | 0.903 | 0.426 | 0.482 | 0.959 | 1.131 |
| NUN-UTD | 0.979 | 0.534 | **0.999** | 1.444 | 1.871 | 0.975 | 0.959 | **0.984** | 0.999 | 1.025 |
| HUTDNet | 0.966 | 0.341 | 0.458 | 1.083 | 1.343 | 0.974 | 0.269 | 0.382 | 1.087 | 1.420 |
| PCL-HUTD | **0.997** | 0.062 | 0.792 | **1.728** | **12.837** | **0.995** | 0.229 | 0.809 | **1.576** | **3.538** |

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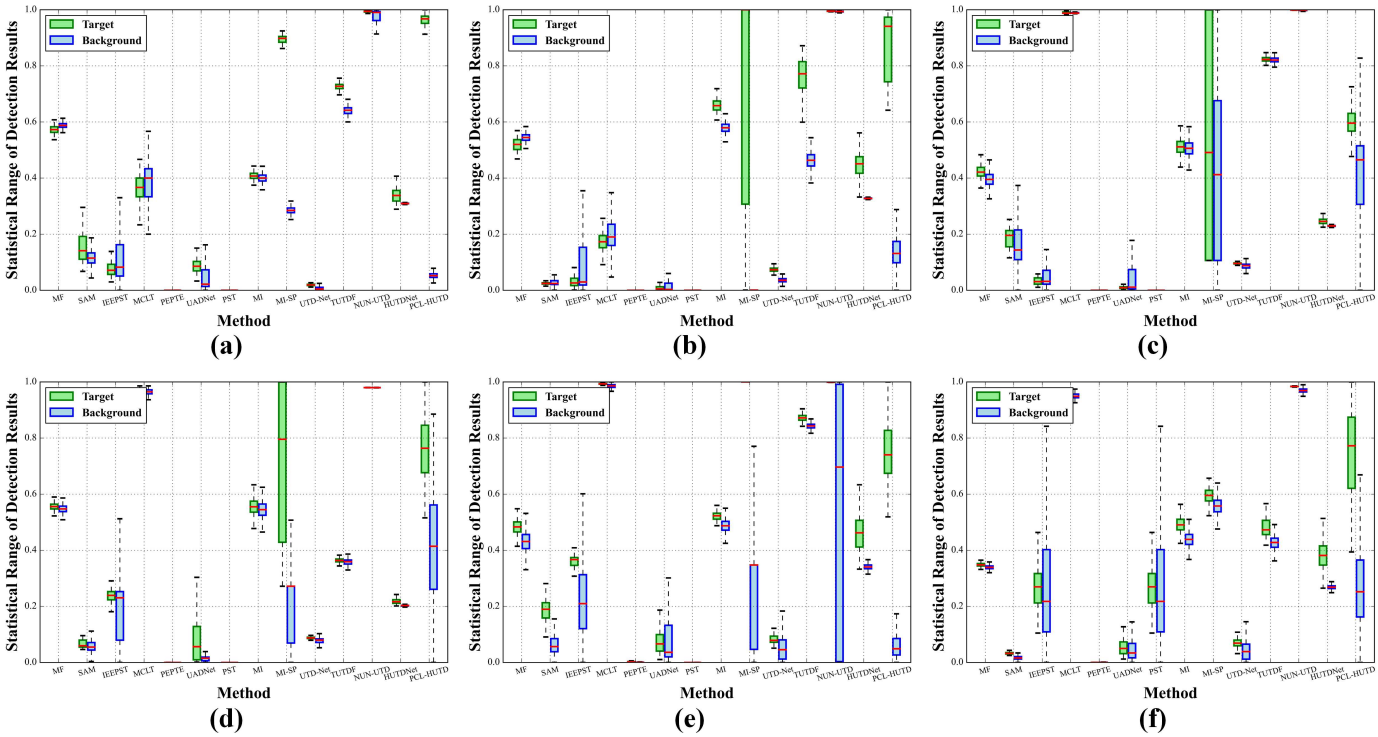



Fig. 7. Statistical distributions of detection scores for target (green) and background (blue) pixels in six subscenes. (a) Lake Scene 1. (b) Lake Scene 2. (c) River Scene 1. (d) River Scene 2. (e) Sea Scene 1. (f) Sea Scene 2. A larger separation between boxplots indicates stronger TBS.

worth noting that methods performing well under a single-evaluation criterion often fail to sustain such performance when assessed holistically. For instance, although MI-SP exhibits relatively high $\mathrm{AUC}_{(D,\tau)}$ scores in several scenes, it also suffers from elevated $\mathrm{AUC}_{(F,\tau)}$ values, leading to poor $\mathrm{AUC}_{\mathrm{SNPR}}$ and excessive background activation. Conversely, classical detectors, such as MF and SAM, maintain low false alarm rates as indicated by small $\mathrm{AUC}_{(F,\tau)}$ values, but they fail to achieve sufficient target recall, resulting in low $\mathrm{AUC}_{(D,\tau)}$ and ultimately suboptimal $\mathrm{AUC}_{\mathrm{OA}}$ scores. In contrast, PCL-HUTD demonstrates consistently balanced performance across all evaluated metrics. This robustness across different evaluation perspectives further validates its effectiveness in real-world HUTD scenarios.

*4) TBS Maps:* To further evaluate the separability between target and background responses, we conduct a statistical analysis of detection score distributions, as illustrated in Fig. 7. Each subplot corresponds to a specific scene from the ATR2-HUTD benchmark. For each method, green and blue boxplots indicate the distributions of detection scores for target and background pixels, respectively. An ideal detector is expected to produce high and compact score distributions for targets, while maintaining low and narrowly distributed background responses, thereby minimizing the overlap between the target and background distributions.

The proposed PCL-HUTD consistently demonstrates the most favorable separation across all six scenes. In both Lake scenes [Fig. 7(a) and 7(b)], it produces tightly concentrated background scores near zero, while the target responses are not only higher in magnitude but also exhibit minimal variance. This pronounced separation strongly correlates with the superior ROC and AUC results reported earlier. In comparison, methods, such as MI-SP and NUUTD, occasionally generate elevated target responses [e.g., scenes Fig. 7(b) and 7(e)], but their background distributions are also broad or skewed toward higher values. This leads to increased overlap between target and background scores, reducing practical discriminability and increasing the risk of false positives. Unsupervised baselines, such as UADNet and PST, show heavily overlapped target-background distributions, confirming their low discriminability and explaining the poor recall observed in earlier ROC analyses. In the Sea scenes [Fig. 7(e) and (f)], where spectral interference from structures and water shadows is prevalent, many approaches (e.g., MCLT and IEEPST) struggle to maintain clear class boundaries, as evidenced by substantial overlap in their boxplots. In contrast, PCL-HUTD preserves strong contrast between target and background responses, reflecting robust feature characterization under spectral ambiguity. The River scenes [Fig. 7(c) and (d)] present greater challenges due to complex textures and weaker target visibility. Physics-informed methods, like PEPTE and HUTDNet, suffer from fragmented targets or broadened background responses in these complex underwater environments. Even under these adverse conditions, PCL-HUTD achieves well-separated score distributions with clear statistical margins, indicating strong generalization capability across diverse environments. Notably, classical methods, such as SAM and SOTA, and approaches, like UTD-Net, yield low and highly overlapping response distributions, highlighting their limited discriminative capacity in complex underwater scenarios.

Overall, this TBS-based analysis complements the findings from ROC evaluations, offering further evidence that PCL-

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TABLE IV
QUANTITATIVE RESULTS OF ABLATION STUDIES ON THE ATR2-HUTD DATASET

| Module Name | Design | $AUC_{(D,F)}\uparrow$ | $AUC_{(F,\tau)}\downarrow$ | $AUC_{(D,\tau)}\uparrow$ | $AUC_{OA}\uparrow$ | $AUC_{SNPR}\uparrow$ |
|---|---|---|---|---|---|---|
| **PCL-HUTD** | N/A | 0.986 | 0.206 | 0.808 | 1.588 | 6.272 |
| **PITC Module** | w/o Physics Informed Modeling | 0.876 | 0.317 | 0.754 | 1.313 | 2.379 |
| | w/o Nonlinear Embedding | 0.765 | 0.375 | 0.706 | 1.096 | 1.883 |
| **HSGTC Module** | w/o Hard Sample Mining | 0.916 | 0.274 | 0.798 | 1.440 | 2.912 |
| | w/o HSGTC module | 0.694 | 0.432 | 0.649 | 0.911 | 1.502 |
| **PCRS Module** | w/o PCRS module | 0.563 | 0.478 | 0.518 | 0.603 | 1.084 |
| **CL Paradigm** | w/o CL Paradigm | 0.752 | 0.383 | 0.719 | 1.088 | 1.877 |

HUTD not only delivers high detection accuracy but also maintains interpretable and stable response behavior, which is an essential quality for real-world HUTD applications.

### D. Ablation Study

To investigate the contribution of each core component in the proposed PCL-HUTD framework, we conduct a comprehensive ablation study. Unlike conventional designs that directly remove individual modules, we adopt a more practical without-style disassembly strategy. This choice is motivated by the fact that PCL-HUTD forms a tightly integrated pipeline, where the output of each module is indispensable for the functioning of the subsequent stages. Consequently, eliminating a module in isolation would compromise the entire framework. Instead, we implement targeted ablation variants that deactivate or replace specific algorithmic elements, thereby preserving structural integrity while isolating the impact of individual design choices.

Based on the overall architecture, we categorize PCL-HUTD into four principal modules: physics-informed transformation construction (PITC), HSGTC, prototype contrastive representation selection (PCRS), and the CL paradigm. The corresponding experimental results are summarized in Table IV.

*1) Effect of the PITC Module:* As the PITC module generates the transformed input for downstream learning, it cannot be fully removed. Instead, we evaluate two reduced variants.

1) *Without Physics-Informed Modeling:* This setting discards the modeling of underwater optical attenuation and directly uses the reference target spectrum as input for the nonlinear embedding network.
2) *Without Nonlinear Embedding:* This version removes the nonlinear projection network $\mathcal{F}_\theta$ and only performs the physics-informed modeling to achieve a linear transformation of the input spectra as the target characterization.

The results in Table IV show that removing the physics-based modeling significantly reduces $AUC_{(D,F)}$ (from 0.986 to 0.876) and $AUC_{OA}$ (from 1.588 to 1.313), demonstrating the importance of simulating underwater propagation for discriminative input construction. Further omitting the nonlinear embedding leads to a more substantial drop ($AUC_{(D,F)}$ = 0.765 and $AUC_{SNPR}$ = 1.883), confirming the necessity of task-adaptive feature learning beyond raw spectral analysis.

*2) Effect of the HSGTC Module:* We evaluate the role of the hard-sample guided learning strategy through two variants.

1) *Without Hard Sample Mining Strategy:* This variant removes difficulty-based sample selection and applies uniform contrastive learning across all samples.
2) *Without HSGTC Module:* This variant substitutes the contrastive strategy with a traditional MIL approach [27].

Excluding hard-sample mining results in a moderate performance drop ($AUC_{(D,F)}$ = 0.916 and $AUC_{OA}$ = 1.440), indicating that difficulty-aware optimization enhances feature separability. Replacing the entire HSGTC module with MIL causes a sharp degradation ($AUC_{(D,F)}$ = 0.694 and $AUC_{OA}$ = 0.911), underscoring the superiority of contrastive mechanisms over instance-level heuristics in underwater spectral scenarios.

*3) Effect of the PCRS Module:* To assess the effect of clustering results refinement, we examine a variant without this module.

1) *Without PCRS Module:* This configuration directly utilizes clustering outputs for sample characterization, omitting the consistency-guided refinement process.

This variant yields the most substantial degradation across all metrics ($AUC_{(D,F)}$ = 0.563 and $AUC_{OA}$ = 0.603), emphasizing that the PCRS module is critical for eliminating noisy samples and maintaining compact prototype clusters, which are essential for contrastive representation learning.

*4) Effect of the CL Paradigm:* We further investigate the role of progressive sample scheduling via the following configuration.

a) *Without CL Paradigm:* The model is trained using a standard self-supervised strategy without any scheduling based on sample difficulty or reliability.

The absence of CL results in suboptimal generalization and unstable convergence ($AUC_{(D,F)}$ = 0.752 and $AUC_{SNPR}$ = 1.877), confirming the utility of gradually increasing learning complexity in real-world HUTD tasks.

*5) Summary:* The results of this ablation study reveal that each module in PCL-HUTD offers distinct and complementary benefits. In particular, PCRS and HSGTC are most critical for

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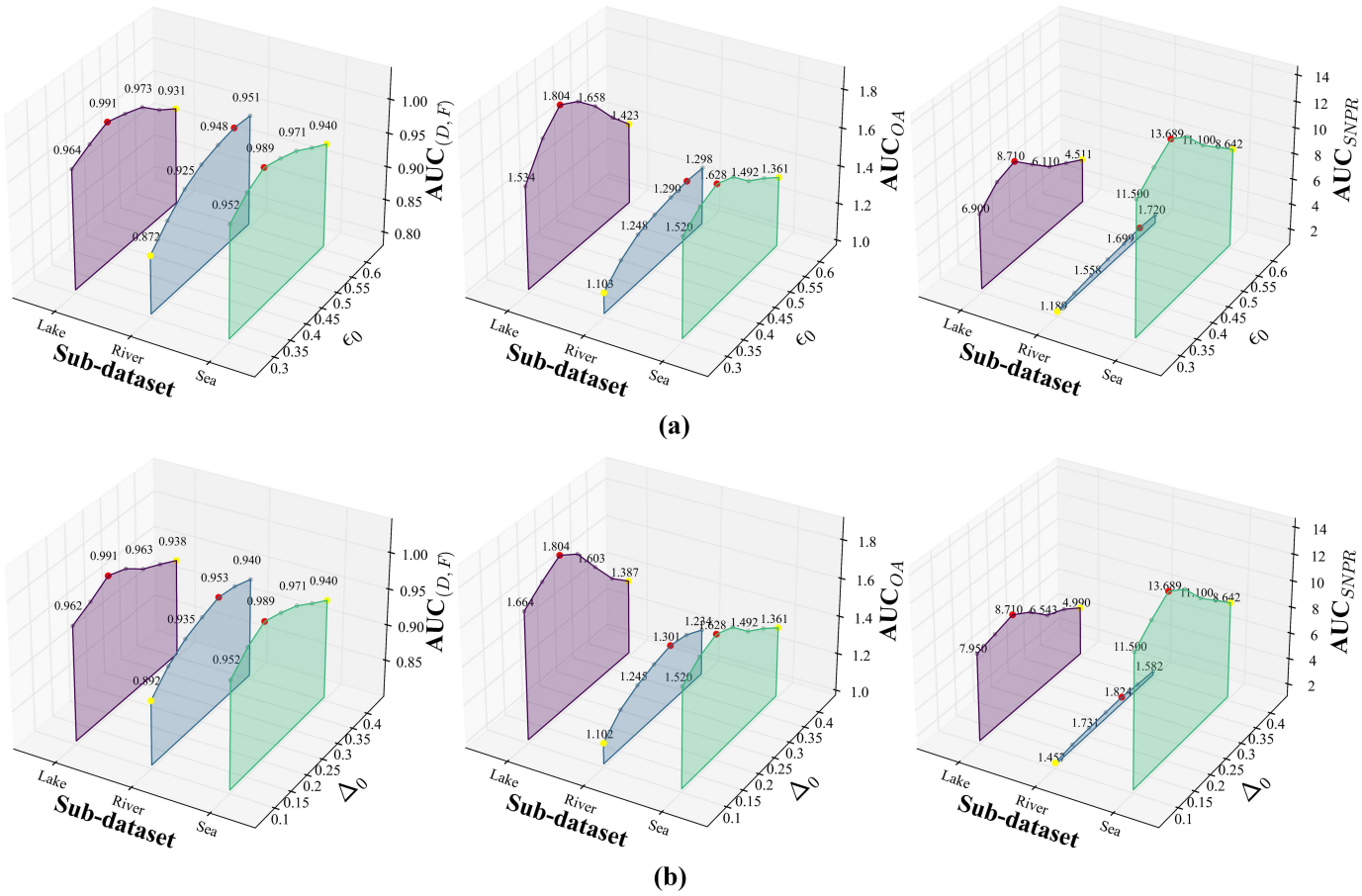



Fig. 8. Hyperparameter analysis results. (a) Effect of $\epsilon_0$. (b) Effect of $\Delta_0$. Three evaluation metrics are plotted: $\mathrm{AUC}_{(D,F)}$, $\mathrm{AUC}_{\mathrm{OA}}$, and $\mathrm{AUC}_{\mathrm{SNPR}}$.

achieving high discriminability and robustness, while PITC and CL provide essential physical consistency and training stability. Their combined effect enables PCL-HUTD to achieve strong performance in complex underwater environments.

### E. Hyperparameter Analysis

In the proposed PCL-HUTD framework, two key hyperparameters play pivotal roles in shaping the quality of physically informed samples and the robustness of contrastive target characterization. These are the base neighborhood threshold $\epsilon_0$ and the initial perturbation radius $\Delta_0$. To validate the empirical settings of these hyperparameters across different scenes, we conduct controlled experiments on the ATR2-HUTD benchmark. In each test, we vary one hyperparameter while fixing the other at its optimal value.

*1) Influence of $\epsilon_0$:* As illustrated in the top row of Fig. 8, $\epsilon_0$ directly determines the spectral proximity criterion used in pixel clustering and prototype formation. Our results reveal that this parameter exhibits distinct scene-specific sensitivity. In the *Lake* and *Sea* scenarios, moderate values, such as $\epsilon_0 = 0.40$ and $\epsilon_0 = 0.50$ yield optimal results, achieving $\mathrm{AUC}_{(D,F)}$ values of 0.991 and 0.989, $\mathrm{AUC}_{\mathrm{OA}}$ values of 1.804 and 1.628, and $\mathrm{AUC}_{\mathrm{SNPR}}$ scores of 8.710 and 13.689, respectively. These configurations strike a favorable balance between representation generalization and background suppression. In contrast, the *River* scene, which is characterized by higher spectral complexity and environmental variability, benefits from a larger threshold of $\epsilon_0 = 0.55$. This setting allows for sufficient cluster diversity to capture subtle spectral variations. Conversely, using a smaller value (e.g., $\epsilon_0 = 0.30$) leads to under-clustering and reduced interclass separability, while excessively large values (e.g., $\epsilon_0 = 0.60$) introduce noisy associations, thereby degrading all performance metrics.

*2) Influence of $\Delta_0$:* The bottom row of Fig. 8 demonstrates the effect of $\Delta_0$ on contrastive sample generation. This parameter controls the perturbation radius in the reliability-aware clustering process, thereby affecting the scale of selected samples. Similar to $\epsilon_0$, the optimal value of $\Delta_0$ varies across scenes. For both *Lake* and *Sea* scenes, a value of $\Delta_0 = 0.2$ is found to be most effective, enabling the model to introduce sufficient spectral perturbation without sacrificing consistency. In these cases, $\mathrm{AUC}_{(D,F)}$ reaches 0.991 and 0.989, respectively. However, in the more complex *River* scenes, the model benefits from a higher degree of perturbation with $\Delta_0 = 0.3$, which enables the capture of more challenging and informative samples amid highly cluttered and dynamic backgrounds. A smaller setting, such as $\Delta_0 = 0.1$, results in insufficient variation, limiting contrastive effectiveness. Conversely, larger values introduce instability in optimization and reduce detection precision.

*3) Summary:* The above-mentioned results and analyses demonstrate that the proposed framework remains both robust and effective when the hyperparameters $\epsilon_0$ and $\Delta_0$ are chosen within reasonable ranges. A careful selection of these

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parameters is not critical for achieving satisfying performance. This indicates that the proposed *PCL-HUTD* framework exhibits low sensitivity to hyperparameter settings, thereby enhancing its practical applicability across diverse underwater environments. To further support real-world deployment, we have developed a scene-adaptive hyperparameter tuning strategy. The details of its implementation are provided in Section V of the Supplementary Material.

## VI. Conclusion

In this article, we have proposed PCL-HUTD, a novel physics-informed CL framework for hyperspectral underwater target detection. The proposed method tackles two fundamental challenges in this domain: spectral distortion induced by water-related propagation effects and high intraclass spectral variability arising from complex aquatic environments. In contrast to conventional separation-based or parameter-sensitive approaches, PCL-HUTD obviates the need for explicit environmental parameter estimation by embedding physical priors into a progressive, self-supervised learning pipeline.

To this end, we introduce a physics-guided transformation module that leverages underwater imaging models to construct physically consistent target representations. This is followed by a hard-sample aware contrastive characterization module that enhances target-background discriminability in a latent embedding space. To enable label-free sample reliability estimation, we develop an unsupervised clustering strategy guided by perturbation consistency across multiple clustering regimes. Furthermore, these components are integrated into a closed-loop CL framework, which dynamically refines the reliability estimation and representation learning processes, thereby ensuring stable convergence and strong generalization under diverse environmental conditions.

Extensive experiments conducted on six real-world HUTD scenarios demonstrate that PCL-HUTD consistently outperforms SOTA methods across multiple metrics, including detection accuracy, robustness to threshold variation, and feature-level separability. The effectiveness of each proposed module is further validated through ablation studies and hyperparameter sensitivity analysis, confirming the soundness of the overall design.

In summary, this work presents a unified and physically aware solution to HUTD. By harmonizing physical modeling, discriminative learning, and adaptive curriculum scheduling, PCL-HUTD provides a robust and deployable framework for real-world oceanic sensing applications.

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